MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, RANDOM ACCESS, KEY CODE ZJT

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic random access memory module.

* 1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 3.242 failures per million operating hours for class I or 10.869 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ZJT.

3.5.2 Size. The module size shall be 1A.

* 3.5.3 Weight. The module shall have a maximum weight of 45 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/303-1 for class I and M28787/303-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

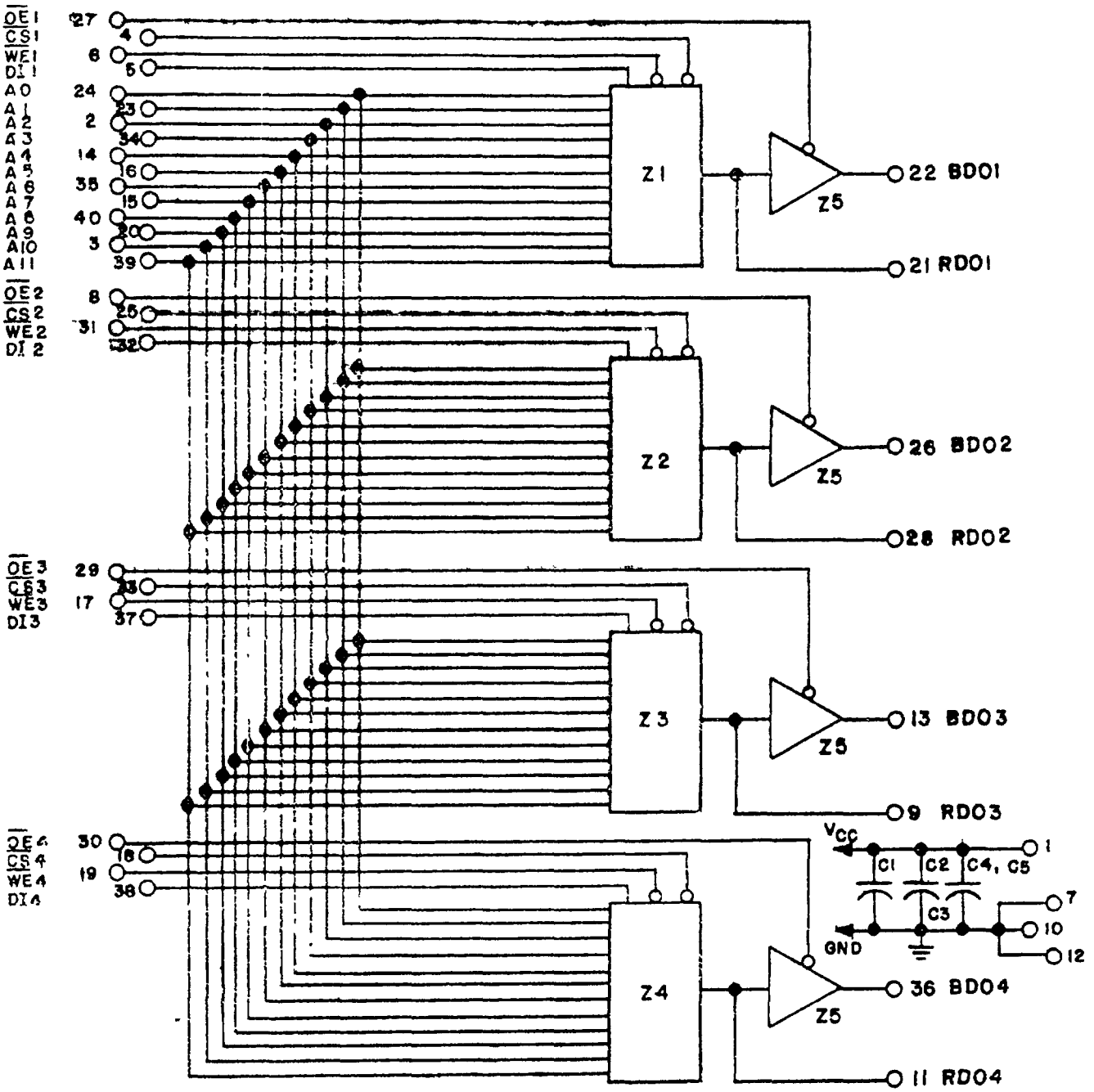
3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

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1. RDO1-RDO4 Ram data outputs.
2. BDO1-BDO4 Buffered data outputs.
3. DI1-DI4 Data inputs.
4. $\overline{OE}$1-$\overline{OE}$4 Output enable inputs (buffer).
5. $\overline{CS}$1-$\overline{CS}$4 Circuit select inputs.
6. $\overline{WE}$1-$\overline{WE}$4 Write enable inputs.
7. A0-A11 Address inputs.

* FIGURE 1. ZJT module function diagram, random access memory.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ -3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} stress	IV	Q	← See test table IV →										
C1-C24	Input leakage current	V	X	0	10	0	10	0	10	0	10	0	10	µA
C25-C28	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D4	Output leakage current, low level outout	VI	X	-1.65	0.05	-1.65	0.05	-1.65	0.05	-1.65	0.05	-1.65	0.05	mA
D5-D8	Output leakage current, low level output	VI	X	-40	40	-40	40	-40	40	-40	40	-40	40	µA
E1-E4	Output leakage current, high level output	VI	X	-90	90	-90	90	-90	90	-90	90	-90	90	µA
E5-E8	Output leakage current, high level output	VI	X	-40	40	40	40	-40	40	-40	40	-40	40	µA
F1-F4	Output short circuit current	VII	X	-30	-70	-30	-70	-30	-70	-30	-70	-30	-70	mA
F5-F8	Output short circuit current	VII	X	-30	-250	-30	-250	-30	-250	-30	-250	-30	-250	mA
G1-G4	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
H1-H8	Output voltage, low level	IX	X	10	400	10	400	10	400	10	400	10	400	mV
J1-J8	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K4	Input current, high level	X	X	0	40	0	40	0	40	0	40	0	40	µA
L1-L24	Input current, low level	XI	X	-10	1	-10	1	-10	1	-10	1	-10	1	µA
L25-L28	Input current, low level	XI	X	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1-M4	Address input threshold check, low level	XII	X	0	400	0	400	0	400	0	400	0	400	mV
M5-M8	Address input threshold check, high level	XII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
N1-N3	Ground pin continuity	XIII	X	0	1	0	1	0	1	0	1	0	1	Ω
N4-N6	Ground pin continuity	XIII	Q	0.0	0.1	0 0	0 1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
P1	Power supply current	XIV	X	50	774	50	774	50	774	50	774	50	774	mA
P2	Power supply current	XIV	X	20	174	20	174	20	174	20	174	20	174	mA
P3-P6	Power supply current	XIV	X	50	324	50	324	50	324	50	324	50	324	mA
Q1-Q8	Transition time (t_{TLH}), C_L = 50 pF	XV	S	1	10	1	10	1	10	1	12	1	12	ns
Q9-Q16	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	1	13	1	13	1	13	1	15	1	15	ns
Q17-Q24	Transition time (t_{THL}), C_L = 50 pF	XV	S	1	7	1	7	1	7	1	10	1	10	ns
Q25-Q32	Transition time (t_{THL}), C_L = 150 pF	XV	Q	1	10	1	10	1	10	1	13	1	13	ns
R1-R8	Dynamic test, C_L = 50 pF	XVI	X	← No failures (see table XVI) →										
R9-R16	Dynamic test, C_L = 150 pF	XVI	Q	← No failures (see table XVI) →										
S1-S8	Circuit select recovery time (t_{SRL})	XVII	S	5	40	5	40	5	40	5	40	5	40	ns
S9-S16	Circuit select recovery time (t_{SRH})	XVII	S	5	40	5	40	5	40	5	40	5	40	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T1-T4	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	5	18	5	18	5	18	5	20	5	20	ns
T5-T8	Propagation delay time (t_{PLH}), C_L = 150 pF	XVIII	Q	5	28	5	28	5	28	5	30	5	30	ns
T9-T12	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	5	18	5	18	5	18	5	20	5	20	ns
T13-T16	Propagation delay time (t_{PHL}), C_L = 150 pF	XVIII	Q	5	28	5	28	5	28	5	30	5	30	ns
U1-U8	Input crosstalk, low level	XIX	Q	0	500	0	500	0	500	0	500	0	500	mV
U9-U16	Input crosstalk, high level	XIX	Q	0	500	0	500	0	500	0	500	0	500	mV
V1	Power supply transient amplitude	XX	Q	0	600	0	600	0	600	0	600	0	600	mV
V2	Power supply ripple amplitude	XX	Q	0	700	0	700	0	700	0	700	0	700	mV
W	Power supply filter capacitance	XXI	S	6	10	6	10	6	10	6	10	6	10	µF
X1-X4, X17-X28	Input capacitance	XXII	Q	2	15	2	15	2	15	2	15	2	15	pF
X5-X16	Input capacitance	XXII	Q	2	45	2	45	2	45	2	45	2	45	pF
Y1-Y8	Output capacitance	XXIII	Q	2	25	2	25	2	25	2	25	2	25	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C1-C24 and D through W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C1-C24 and D through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C1-C24 and D through Y shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6 NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

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6.3 Input clamp diode voltage. This module has input clamp diodes on the $\overline{OE1}$-$\overline{OE4}$ inputs only. All other inputs have protection but not in the form of clamp diode.

6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.5 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table 1.

	Class I	Class II	Unit
Failure rate	3.242	10.869	Failures/10^6 hours maximum
Power dissipation (all selected)	4.26 max	4.26 max	W
(all deselected)	0.96 max	0.96 max	W
Supply voltage	5.0 ±10 percent	5.0 ±10 percent	V dc
Input voltage			
Low level	0.8 max	0.8 max	V
High level	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V
Input current			
$\overline{CS1}$-$\overline{CS4}$, $\overline{WE1}$-$\overline{WE4}$, DI1-DI4, A0-A11			
Low level (V_{IL} = 0 V)	-10 min, 1 max	-10 min, 1 max	µA
(V_{IL} = 0.4 V)	-10 min, 1 max	-10 min, 1 max	µA
High level (V_{IH} = 2.4 V)	10 max	10 max	µA
(V_{IH} = 5.5 V)	10 max	10 max	µA
Input current			
$\overline{OE1}$-$\overline{OE4}$			
Low level (V_{IL} = 0 V)	-0.1 min, -1.73 max	-0.1 min, -1.73 max	mA
(V_{IL} = 0.4 V)	-1.6 max	-1.6 max	mA
High level (V_{IH} = 2.4 V)	0 min, 40 max	0 min, 40 max	µA
(V_{IH} = 5.5 V)	0 min, 1.0 max	0 min, 1.0 max	mA
Static output voltage			
RD01-RD04			
Low level (I_{OL} = 3.4 mA)	0.01 min, 0.4 max	0.01 min, 0.4 max	V
High level (I_{OH} = -0.96 mA)	2.4 min, V_{CC} max	2.4 min, V_{CC} max	V
BD01-BD04			
Low level (I_{OL} = 16.0 mA)	0.01 min, 0.4 max	0.01 min, 0.4 max	V
High level (I_{OH} = -2.0 mA)	2.4 min, V_{CC} max	2.4 min, V_{CC} max	V
Output drive current			
RD01-RD04			
Low level (V_{OL} = 0.4 V)	3.4 max	3.4 max	mA
High level (V_{OH} = 2.4 V)	-0.96 max	-0.96 max	mA
BD01-BD04			
Low level (V_{OL} = 0.4 V)	16 max	16 max	mA
High level (V_{OH} = 2.4 V)	-2.0 max	-2.0 max	mA
Output leakage current (high impedance state)			
RD01-RD04			
Low level (V_{OL} = 0.4 V)	-1.65 min	-1.65 min	mA
(V_{OL} = 0.4 V)	50 max	50 max	µA
High level (V_{OH} = 2.4 V)	-90 min, 90 max	-90 min, 90 max	µA
BD01-BD04			
Low level (V_{OL} = 0.4 V)	-40 min, 40 max	-40 min, 40 max	µA
High level (V_{OH} = 2.4 V)	-40 min, 40 max	-40 min, 40 max	µA

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**6.5 Module description - Continued.

	Class I	Class II	Unit
Setup time (chip deselect time, t_{SS})	10 min	10 min	ns
(address, t_{AS})	20 min	20 min	ns
(circuit select, t_{SS})	70 min	70 min	ns
(data, t_{DS})	40 min	40 min	ns
Hold time (address, t_{AH})	15 min	15 min	ns
(circuit select, t_{SH})	10 min	10 min	ns
(data, t_{DH})	10 min	10 min	ns
Write pulse width (t_{WP})	50 min	50 min	ns
Address access time (t_{AA})			
Address to RAM data output			
C_L = 50 pF	105 max	105 max	ns
C_L = 150 pF	115 max	115 max	ns
Address to buffer data output			
C_L = 50 pF	125 max	125 max	ns
C_L = 150 pF	135 max	135 max	ns
Circuit select access time (t_{SA})			
Circuit select to RAM data out			
1/ t_{SA1} C_L = 50 pF	90 max	90 max	ns
C_L = 150 pF	100 max	100 max	ns
2/ t_{SA2} C_L = 50 pF	105 max	105 max	ns
C_L = 150 pF	115 max	115 max	ns
Enable to buffer data output (t_{SA})			
Circuit select to RAM data out			
1/ t_{SA1} C_L = 50 pF	110 max	110 max	ns
C_L = 150 pF	120 max	120 max	ns
2/ t_{SA2} C_L = 50 pF	125 max	125 max	ns
C_L = 150 pF	135 max	135 max	ns
Circuit select recovery time (t_{SRL} and t_{SRH})			
RAM data output (C_L < 10 pF)	5 min, 40 max	5 min, 40 max	ns
Buffer data output (C_L < 10 pF)	5 min, 40 max	5 min, 40 max	ns
Write recovery time (t_{WR})			
RAM data output			
C_L = 50 pF	50 max	50 max	ns
C_L = 150 pF	60 max	60 max	ns
Buffer data output			
C_L = 50 pF	70 max	70 max	ns
C_L = 150 pF	80 max	80 max	ns
Propagation delay time (t_{PLH} = t_{PHL})			
RDO outputs to BDO outputs			
C_L = 50 pF	5 min, 20 max	5 min, 20 max	ns
C_L = 150 pF	5 min, 30 max	5 min, 30 max	ns

1/ Circuit deselected for greater than 55 ns prior to selection.

2/ Circuit deselected for a finite time that is less than 55 ns prior to selection.

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* 6.6 Module function. This module contains four random access memories of 4096 words by 1 bit with three-state outputs and three-state buffers.

6.6.1 Input to output relationship. The following describes the essential RAM operation (see table II).

TABLE II. Input to output relationship.

INPUTS				OUTPUTS		MODE
$\overline{OE}$	$\overline{CS}$	$\overline{WE}$	DI	RDO	BDO	
H	X	X	X	X	OFF	Buffer high impedance
L	H	X	X	OFF	X	RAM not selected
L	L	L	L	OFF	X	Write "L"
L	L	L	H	OFF	X	Write "H"
L	L	H	X	DI(t_N)	DI(t_N)	Read

6.6.1.1 Read. The memory is addressed with A0 - All inputs which select one of 4096 words. The memory is enabled by making circuit select low. If the write enable ($\overline{WE}$) is high and the memory is enabled, the stored data is read out on the data out pin. The data read out is the same polarity as the data written in during the write cycle.

6.6.1.2 Write. The memory is addressed and enabled as in the read cycle. If the memory write enable ($\overline{WE}$) is low, the data on the data input pin is written into the addressed word of the memory.

6.6.1.3 RAM three-state outputs. The three-state output mode allows each memory to be connected to a common bus with several like memories. The memory outputs can be placed in the high-impedance state by setting any of the memory's enables at a high level or by setting the memory's write enable to a low level.

6.6.1.4 Buffer three-state outputs. The buffer data outputs can be placed in the high-impedance state by setting the appropriate output enable signal to a high level.

6.6.2 Parallel outputs. Bussing is allowed between like outputs of a number of memories. When outputs of several memories are bussed together, all address lines should be addressed and all output enable lines should be enabled using synchronous clocking.

6.6.2.1 Storage and handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedure should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.

b. Solder-iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

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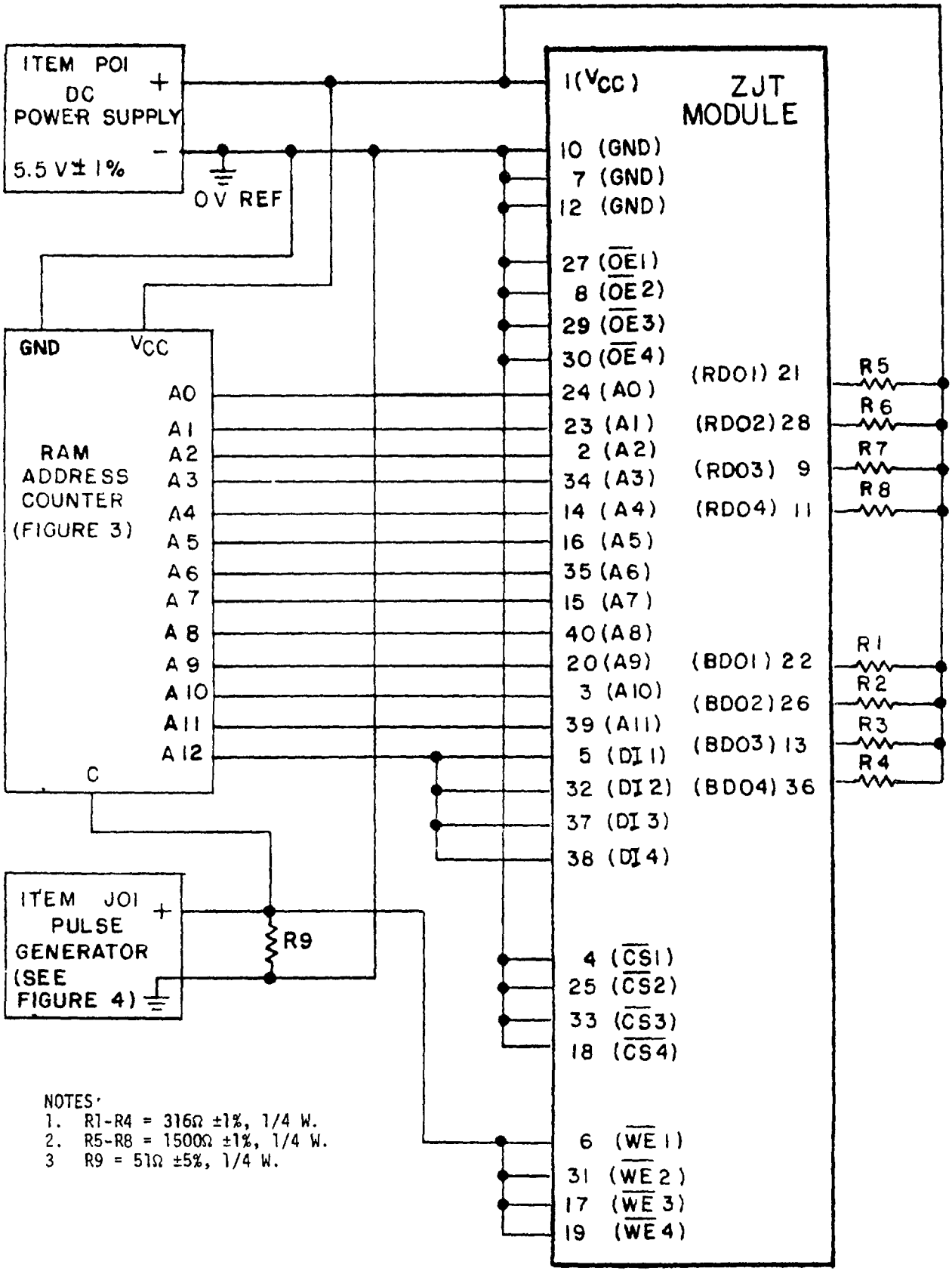
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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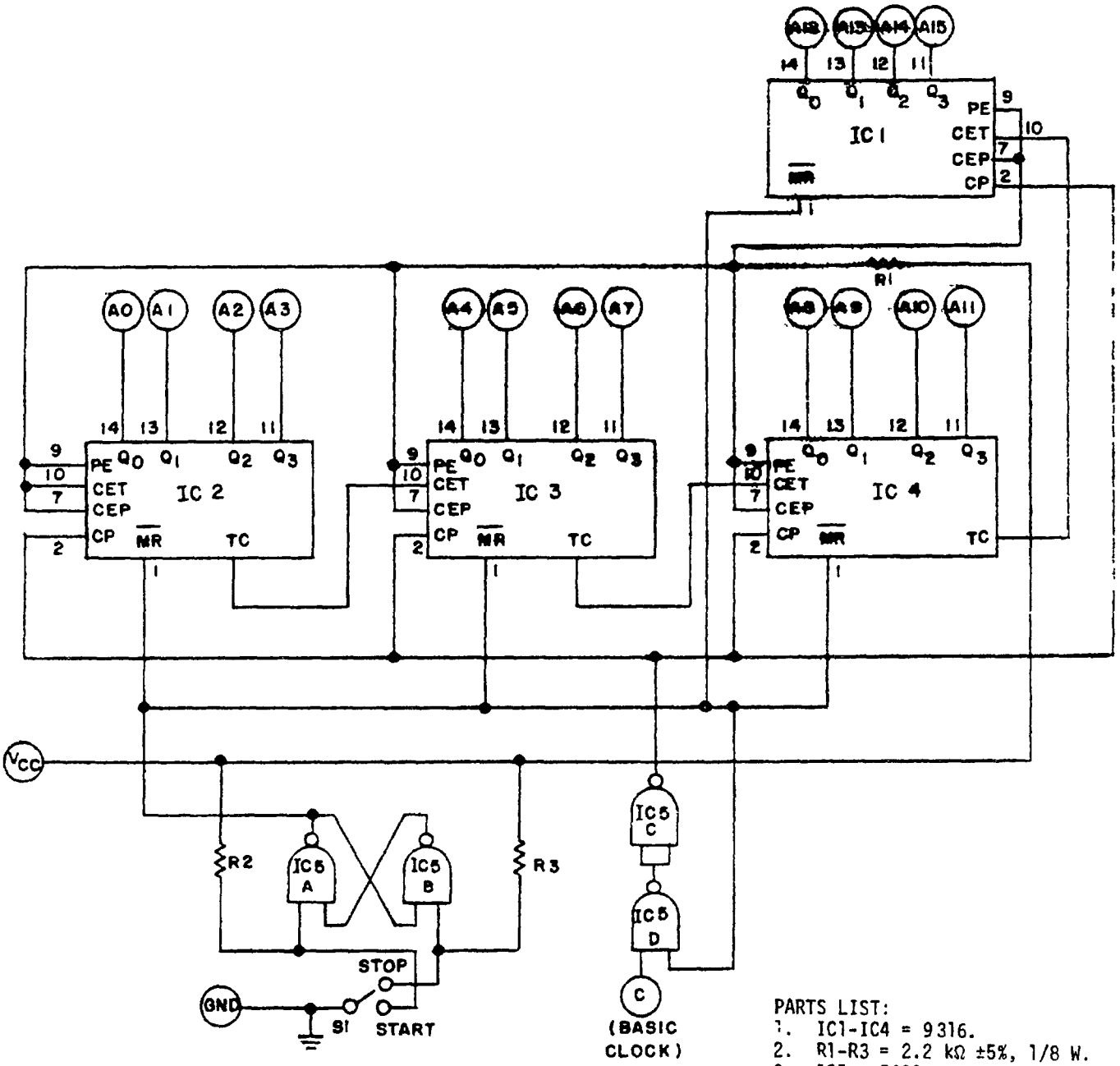
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NOTES:
1. R1-R4 = 316Ω ±1%, 1/4 W.
2. R5-R8 = 1500Ω ±1%, 1/4 W.
3 R9 = 51Ω ±5%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

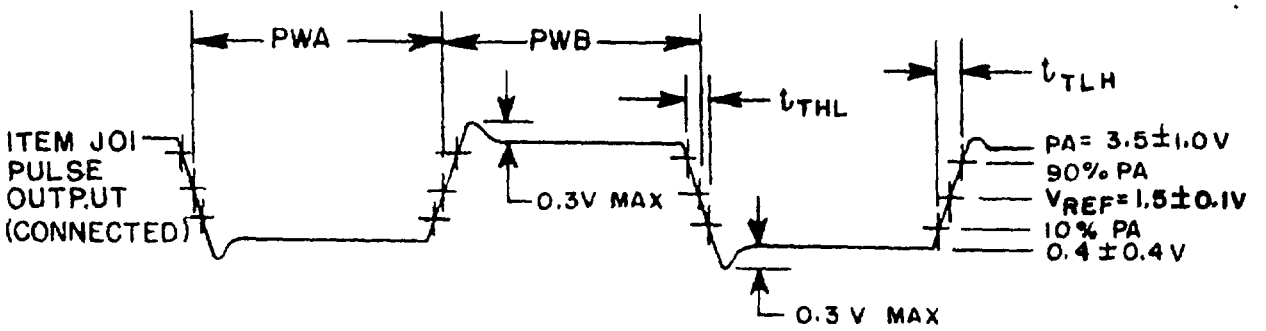
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PARTS LIST:
1. IC1-IC4 = 9316.
2. R1-R3 = 2.2 kΩ ±5%, 1/8 W.
3. IC5 = 5400.
4. S1 = SPDT switch.

FIGURE 3. RAM address counter.

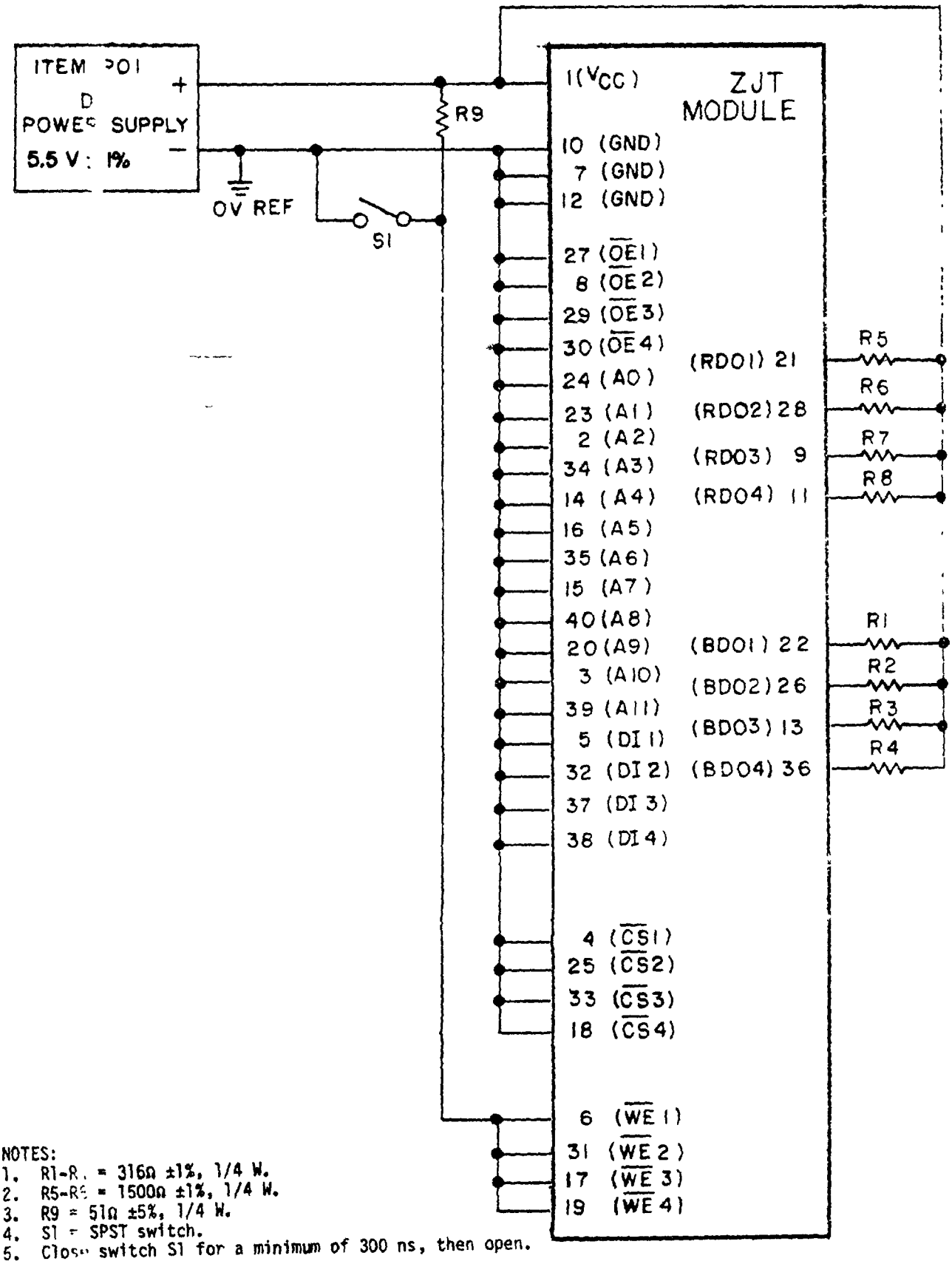
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NOTE Item J01 pulse parameters:
$t_{TLH} = t_{THL} = 30 \pm 20$ ns.
PWA = 250 ±25 ns.
PWB = 300 ±30 ns.

* FIGURE 4. Operating temperature and life test circuit waveform for figure 2.

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NOTES:
1. R1-R4 = 316Ω ±1%, 1/4 W.
2. R5-R8 = 1500Ω ±1%, 1/4 W.
3. R9 = 51Ω ±5%, 1/4 W.
4. S1 = SPST switch.
5. Close switch S1 for a minimum of 300 ns, then open.

FIGURE 5. Component temperature test circuit.

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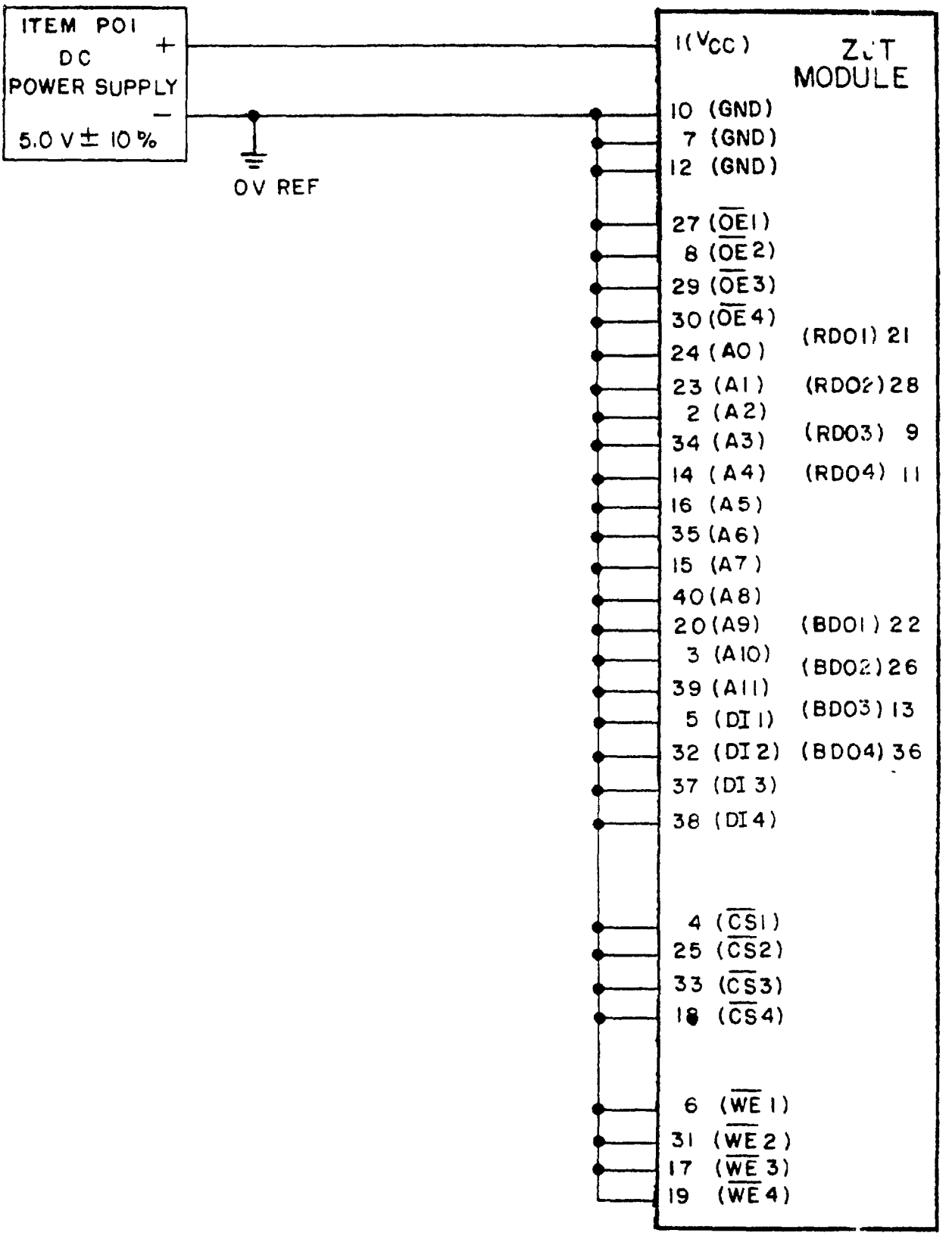


FIGURE 6. Vibration test circuit.

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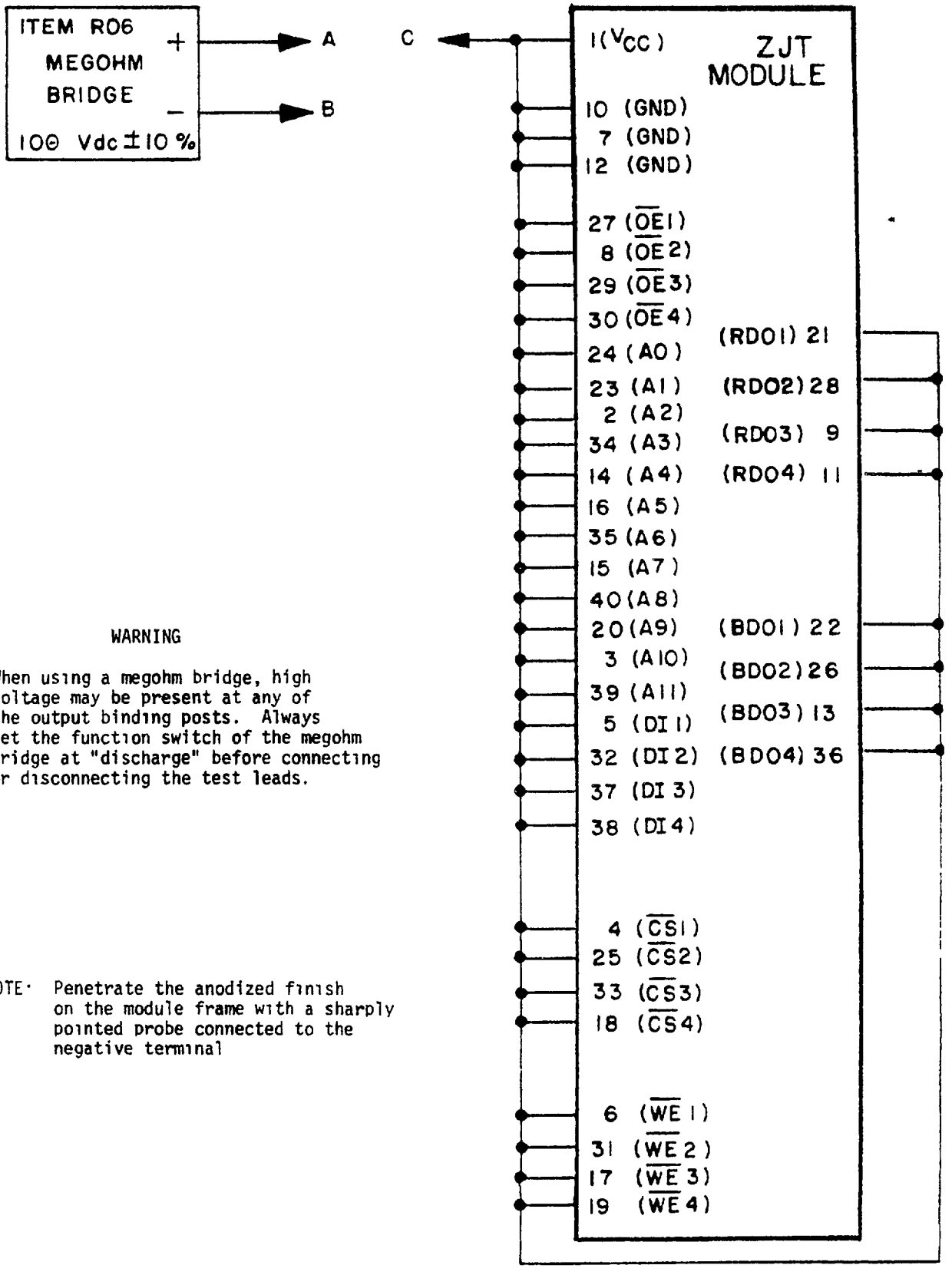
* TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	7

TABLE IV. V_{CC} stress test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
B	V_{CC} stress	← Must pass remainder of tests →			8

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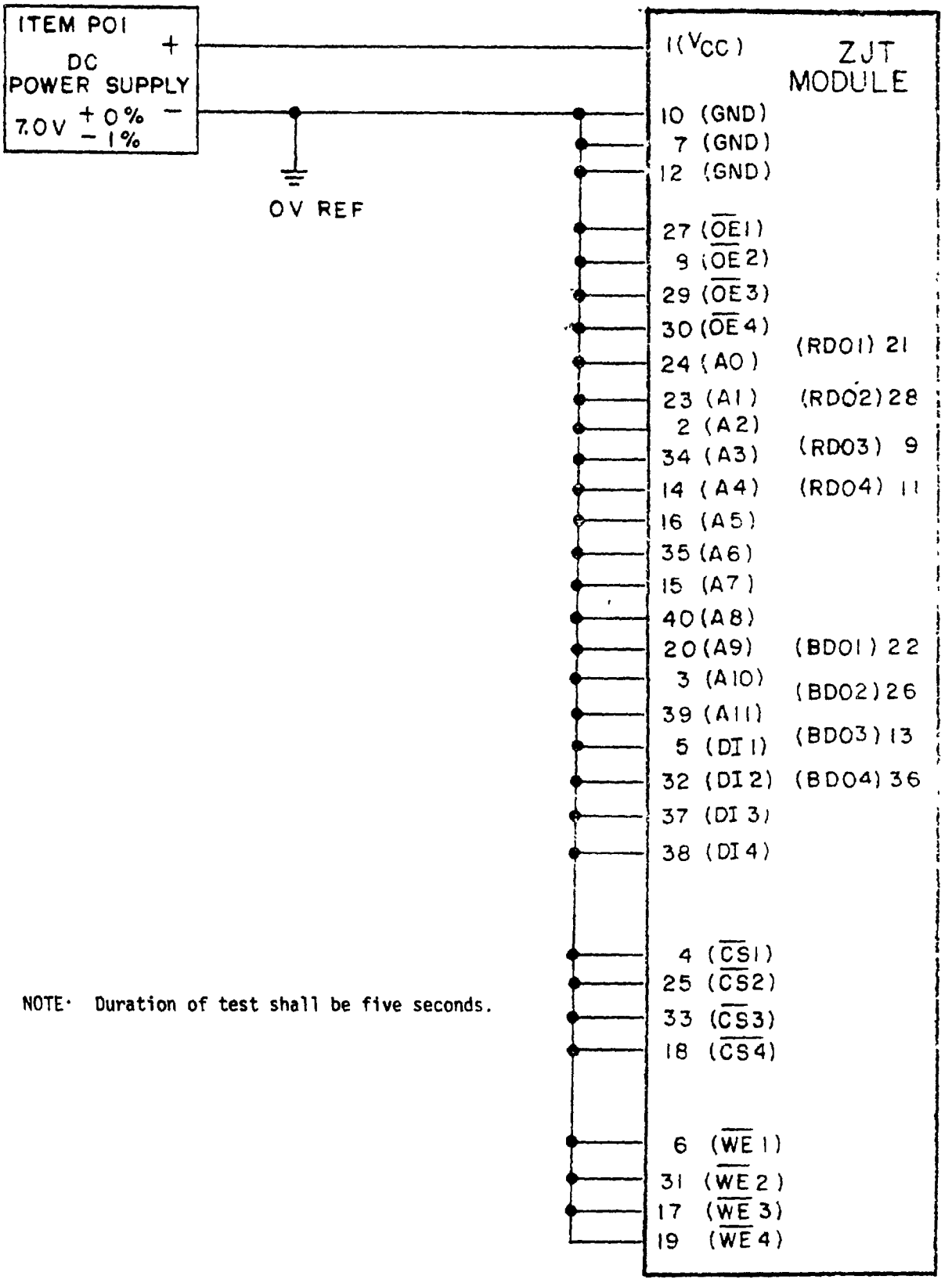
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 7. Isolation test circuit for table III.

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NOTE: Duration of test shall be five seconds.

FIGURE 8. V_{CC} stress test circuit for table IV.

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TABLE V. Input leakage current and input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	$\overline{CS}1$	Input leakage current	2,5,6,14,15,16,17,3,18,19,20,23, 24,25,31,32,33,34,35,37,38,40, 39,27,8,29,30	4	0	10	μA	9
C2	$\overline{CS}2$		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,31,32,33,34,35,37,38,40, 39,27,8,29,30	25				
C3	$\overline{CS}3$		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,31,32,34,35,37,38,40, 39,27,8,29,30	33				
C4	$\overline{CS}4$		2,4,5,6,14,15,16,17,3,19,20,23, 24,25,31,32,33,34,35,37,38,40, 39,27,8,29,30	18				
C5	A0		4,5,6,14,15,16,17,3,18,19,20,23, 25,31,32,33,34,35,37,38,40,39, 27,8,29,30,2	24				
C6	A1		2,4,5,6,14,15,16,17,3,18,19,20, 24,25,31,32,33,34,35,37,38,40, 39,27,8,29,30	23				
C7	A2		4,5,6,14,15,16,17,3,18,19,20,23, 25,31,32,33,34,35,37,38,40,39, 27,8,29,30,24	2				
C8	A3		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,31,32,33,37,38,40,39, 35,8,27,29,30	34				
C9	A4		2,4,5,6,15,16,17,3,18,19,20,23, 24,25,31,32,33,34,35,37,38,39, 40,8,27,29,30	14				

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TABLE V. Input leakage current and input breakdown current test - Continued.

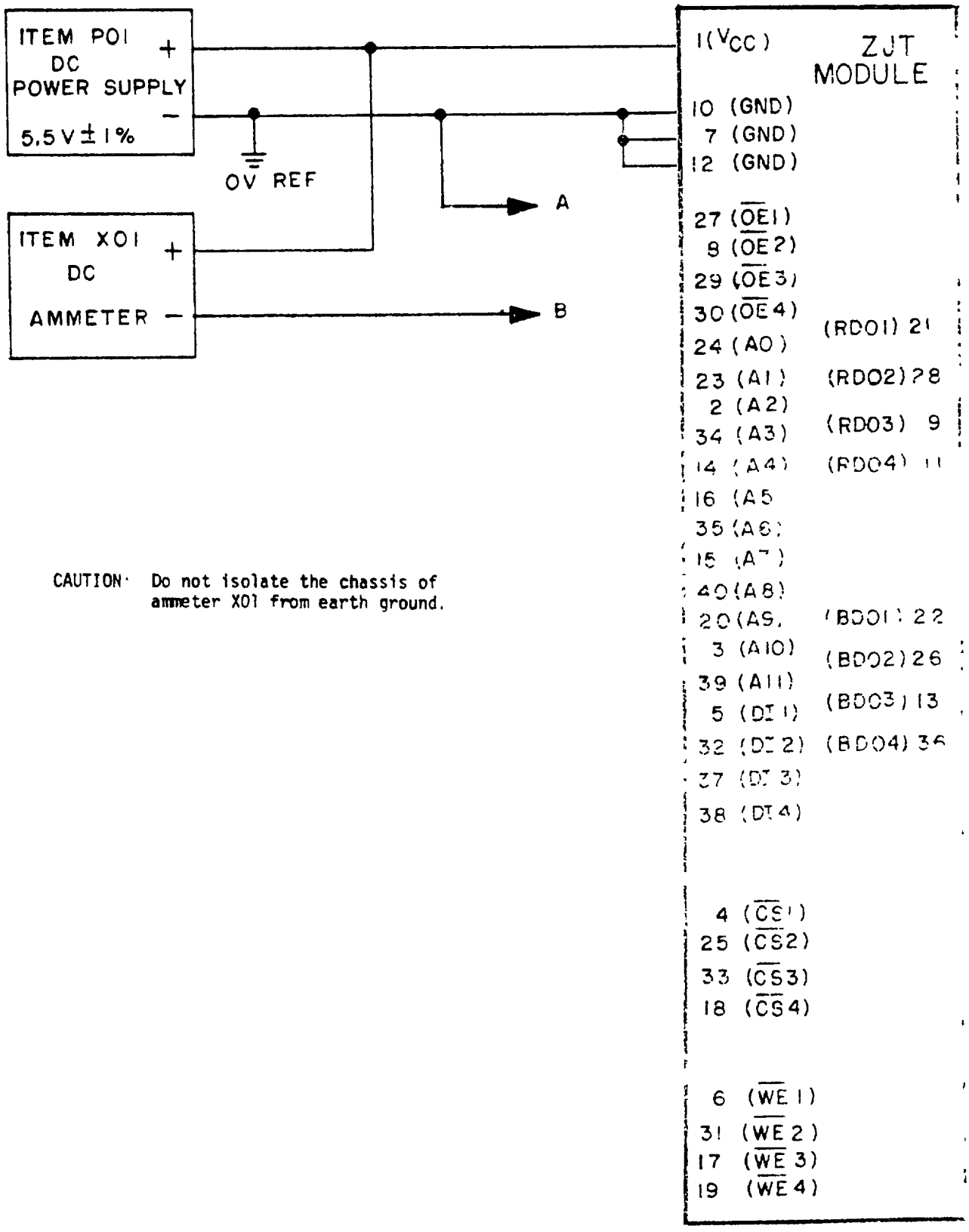
Test condition	Parameter		Connect		Initial limits 25 ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C10	A5	Input leakage current	2,4,5,6,14,15,17,3,18,19,20, 23,24,25,31,32,33,35,37,38,40, 39,34,8,27,29,30	16	0	10	μA	9
C11	A6		2,4,5,6,15,16,17,3,18,19,20,23, 24,25,31,32,33,34,37,38,40,39, 14,8,27,29,30	35				
C12	A7		2,4,5,6,14,16,17,3,18,19,20,23, 24,25,31,32,33,34,35,37,38,40, 39,8,27,29,30	15				
C13	A8		2,4,5,6,14,15,17,3,18,19,20,23, 24,25,31,32,33,34,35,37,38,39, 16,8,27,29,30	40				
C14	A9		2,4,5,6,14,15,16,17,3,18,19,23, 24,25,31,32,33,34,35,37,38,40, 39,8,27,29,30	20				
C15	A10		2,4,5,6,14,15,16,17,18,19,20, 23,24,25,31,32,33,34,35,37,38, 40,39,8,27,29,30	3				
C16	A11		2,3,4,5,6,14,15,16,17,18,19,20, 23,24,25,31,32,33,34,35,37,38, 40,8,27,29,30	39				
C17	$\overline{WE1}$		2,3,4,5,14,15,16,17,3,18,19,20, 23,24,25,31,32,33,34,35,37,38, 40,39,8,27,29,30	6				
C18	$\overline{WE2}$		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,32,33,34,35,37,38,40, 39,8,27,29,30	31				

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TABLE V. Input leakage current and input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25 ±5°C ambient		unit	Test figure
			A to pin	B to pin	Min	Max		
C19	$\overline{WE}3$	Input leakage current	2,4,5,6,14,15,16,3,18,19,20,23, 24,25,31,32,33,34,35,37,38,40, 39,8,27,29,30	17	0	10	µA	9
C20	$\overline{WE}4$		2,4,5,6,14,15,16,17,3,18,20,23, 24,25,31,32,33,34,35,37,38,40, 39,8,27,29,30	19				
C21	DI1		2,4,6,14,15,16,17,3,18,19,20, 23,24,25,31,32,33,34,35,37,38,40, 39,8,27,29,30	5				
C22	DI2		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,31,33,34,35,37,38,40, 39,8,27,29,30	32				
C23	DI3		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,31,32,33,34,35,38,40, 39,8,27,29,30	37				
C24	DI4		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,31,32,33,34,35,37,40, 39,8,27,29,30	38				
C25	$\overline{OE}1$	Input breakdown current	2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,31,32,33,34,35,37,38, 40,39,8,29,30	27		1	mA	
C26	$\overline{OE}2$		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,31,32,33,34,35,37,38, 40,39,27,29,30	8				
C27	$\overline{OE}3$		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,31,32,33,34,35,37,38, 40,39,8,27,30	29				
C28	$\overline{OE}4$		2,4,5,6,14,15,16,17,3,18,19,20, 23,24,25,31,32,33,34,35,37,38, 40,39,8,27,29	30				

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

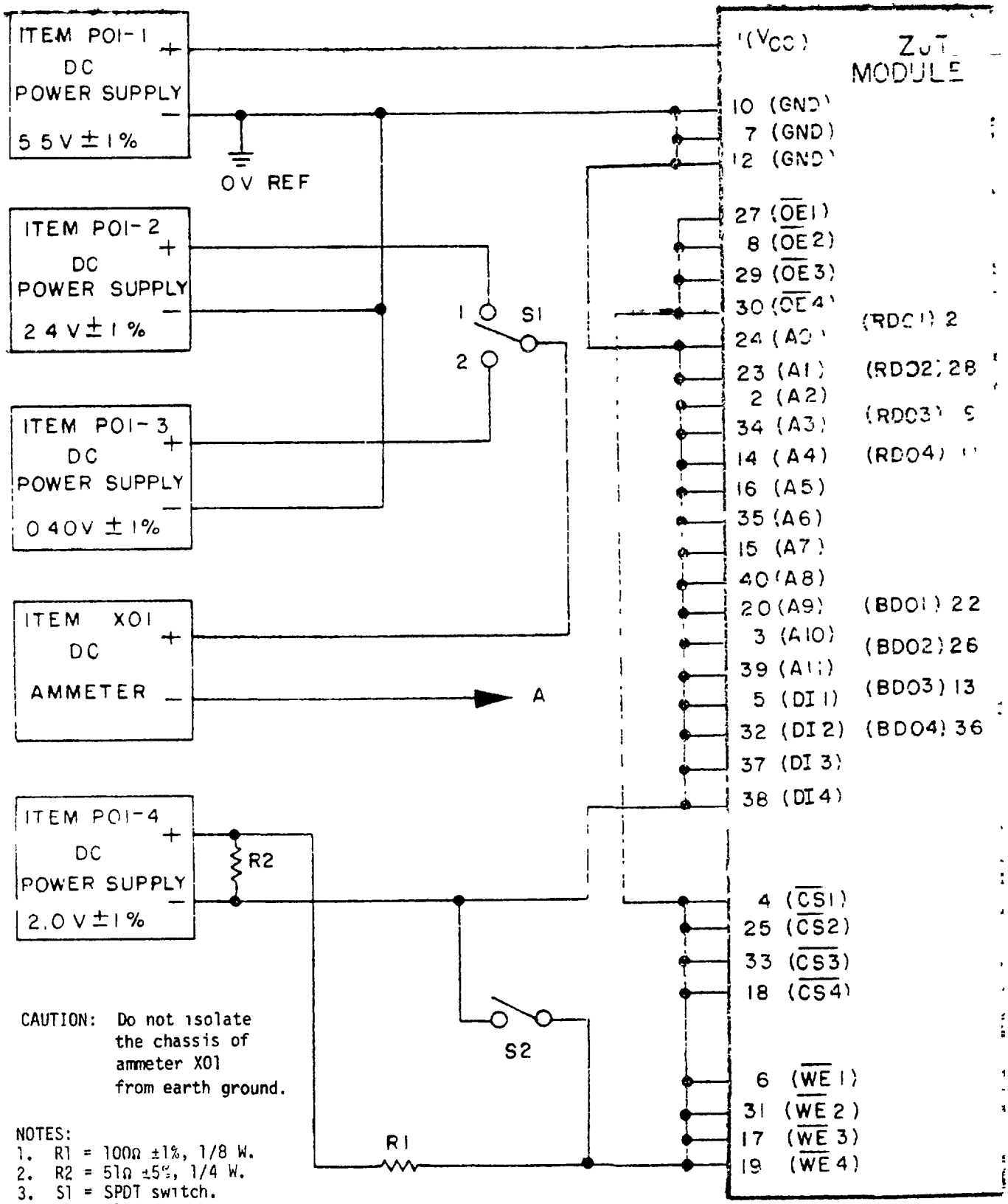
FIGURE 9. Input breakdown current and input leakage current test circuit for table V.

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TABLE VI. Output leakage current (low level and high-level output) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	RD01	Output leakage current, low-level output	21	-1.65	0.05	mA	10
D2	RD02		28				
D3	RD03		9				
D4	RD04		11				
D5	BD01		22	-40	40	µA	
D6	BD02		26				
D7	BD03		13				
D8	BD04		36				
E1	RD01	Output leakage current, high-level output	21	-90	90		
E2	RD02		28				
E3	RD03		9				
E4	RD04		11				
E5	BD01		22	-40	40		
E6	BD02		26				
E7	BD03		13				
E8	BD04		36				

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CAUTION: Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 100Ω ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 1/4 W.
3. S1 = SPDT switch.
4. S2 = SPST switch.
5. Switch S1 in position 2 for test conditions D1-D8.
6. Switch S1 in position 1 for test conditions E1-E8.
7. Before performing a test, close switch S2 for a minimum of 500 m then open and make measurements.

FIGURE 10. Output leakage current (low-level and high-level output) test circuit for table VI.

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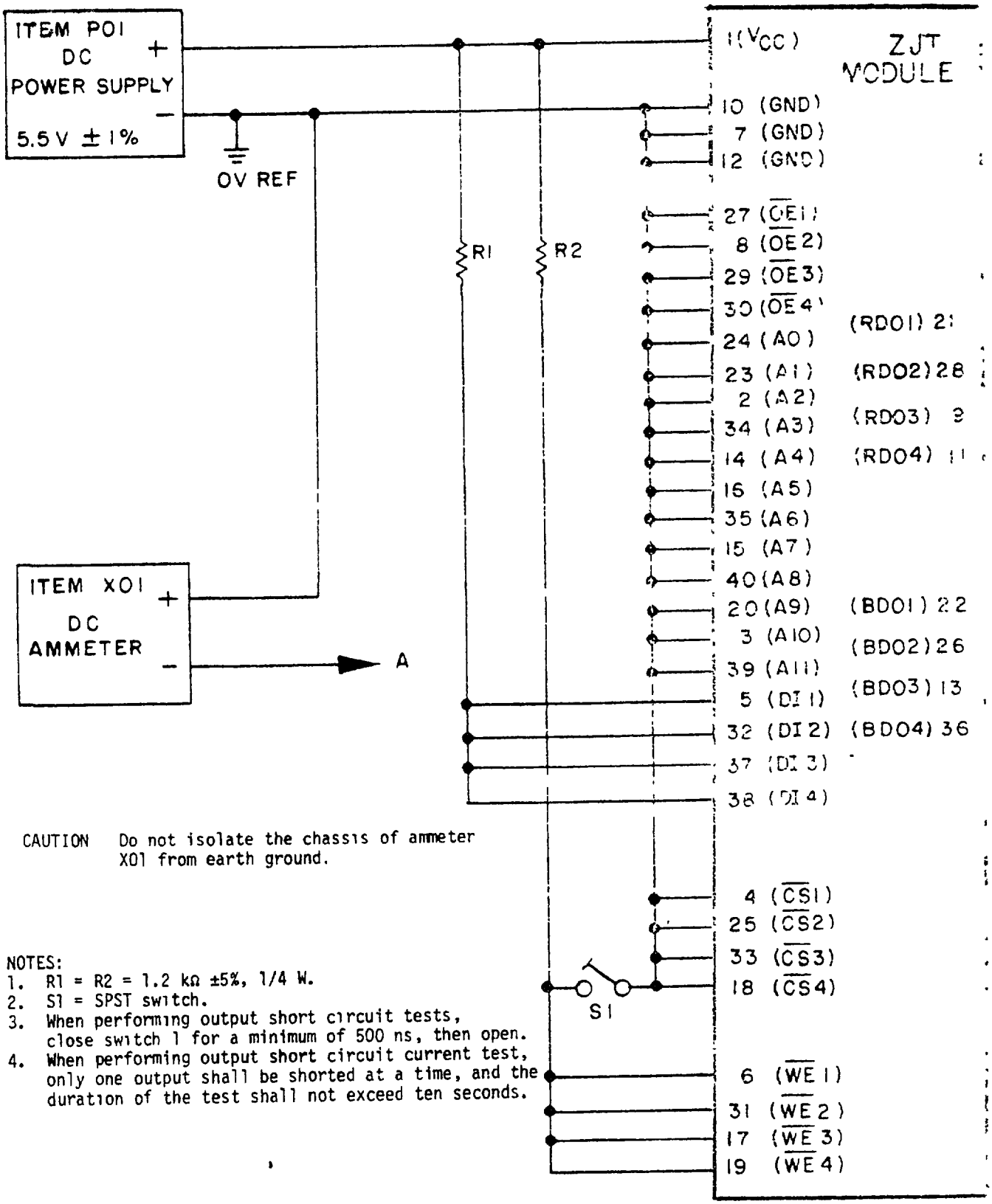
TABLE VII Output short circuit current test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	BDO1	Output short circuit current	22	-30	-70	mA	11
F2	BDO2		26				
F3	BDO3		13				
F4	BDO4		36				
F5	RDO1		21		-250		
F6	RDO2		28				
F7	RDO3		9				
F8	RDO4		11				

TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	$\overline{OE}1$	Input clamp diode voltage	27	-0.4	-1.5	V	12
G2	$\overline{OE}2$		8				
G3	$\overline{OE}3$		29				
G4	$\overline{OE}4$		30				

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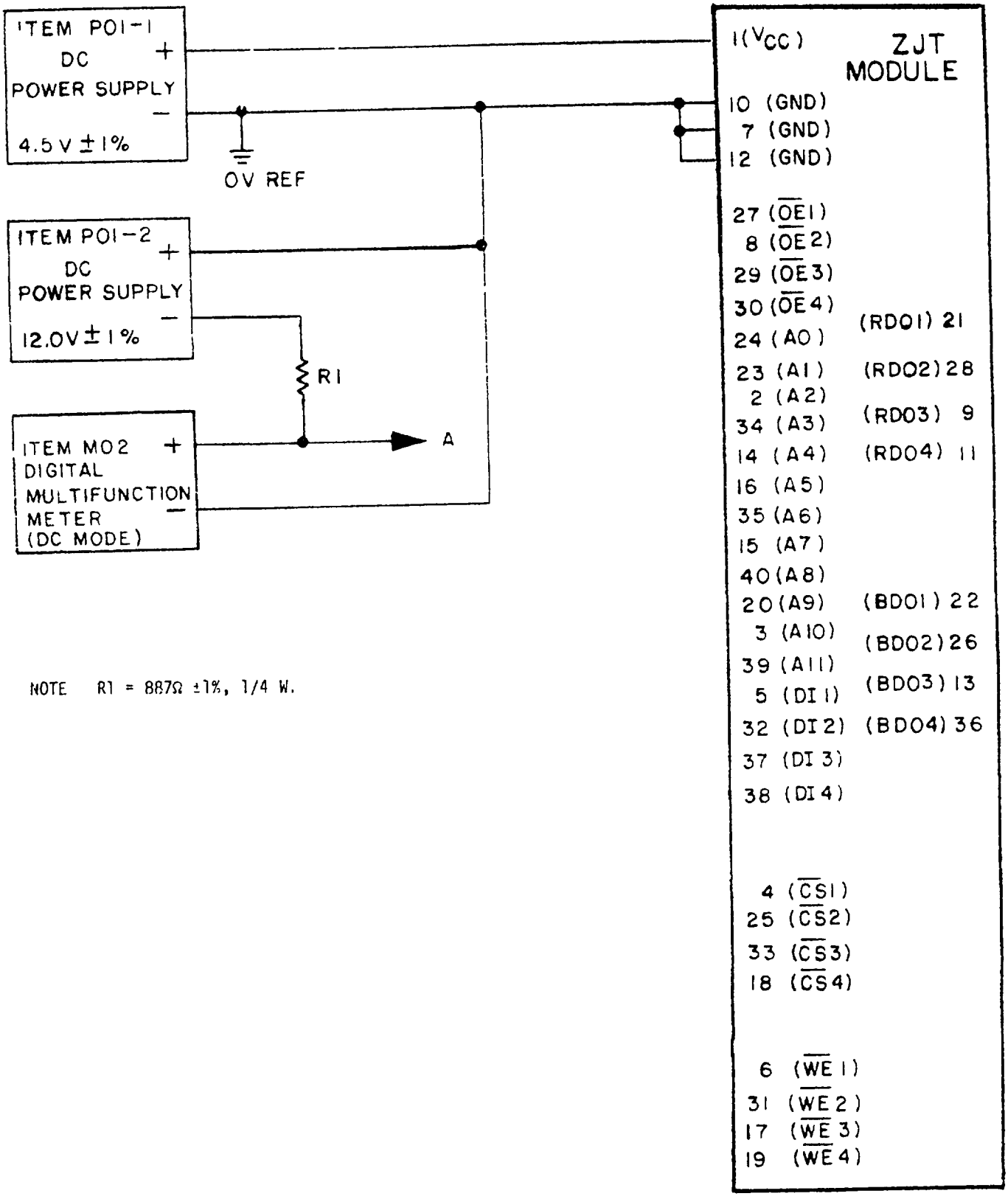


CAUTION Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = R2 = 1.2 kΩ ±5%, 1/4 W.
2. S1 = SPST switch.
3. When performing output short circuit tests, close switch 1 for a minimum of 500 ns, then open.
4. When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed ten seconds.

FIGURE 11. Output short circuit current test circuit for table VII.

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NOTE R1 = 887Ω ±1%, 1/4 W.

FIGURE 12. Input clamp diode voltage test circuit for table VIII.

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* TABLE IX. Output voltage (low-level and high-level) test.

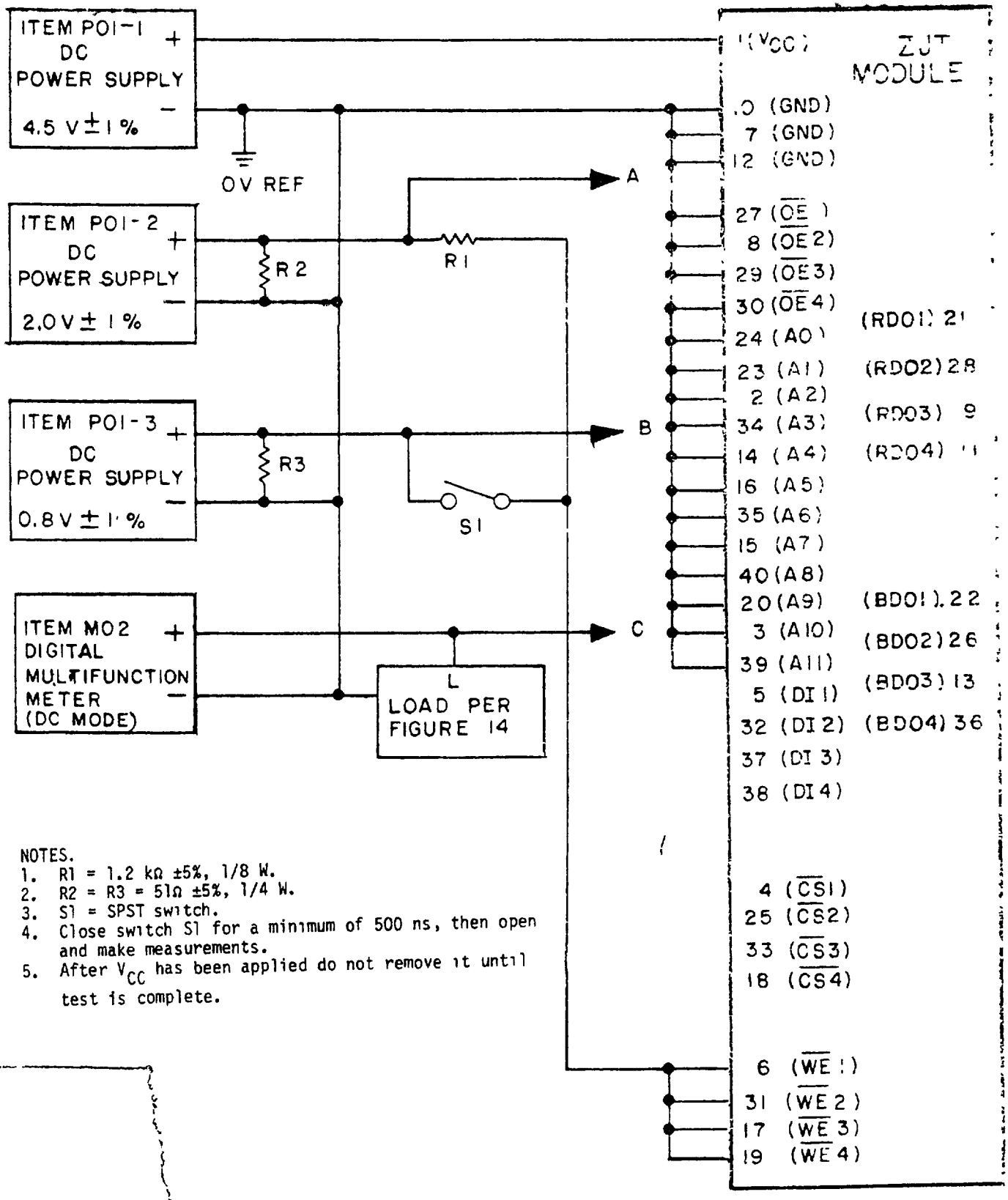
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	RD01	Output voltage, low level	Open	4,5,18, 25,33	21	10	400	mV	13
H2	RD02			4,18,25, 32,33	28				
H3	RD03			4,18,25, 33,37	9				
H4	RD04			4,18,25, 33,38	11				
H5	BD01			4,5,18, 25,33	22				
H6	BD02			4,18,25, 32,33	26				
H7	BD03			4,18,25, 33,37	13				
H8	BD04	▼	▼	4,18,25, 33,38	36	▼	▼	▼	
J1	RD01	Output voltage, high level	5	4,18,25, 32,33,37, 38	21	2.4	4.5	V	
J2	RD02	Output voltage, high level	32	4,5,18, 25,33,37, 38	28	2.4	4.5	V	
J3	RD03	Output voltage, high level	37	4,5,18, 25,32,33, 38	9	2.4	4.5	V	▼

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* TABLE IX. Output voltage (low-level and high-level) test. - Continued

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J4	RD04	Output voltage, high level	38	4,5,18, 25,32,33, 37	11	2.4	4.5	V	13
J5	BD01		5	4,18,25, 32,33,37, 38	22				
J6	BD02		32	4,5,18, 25,33,37, 38	26				
J7	BD03		37	4,5,18, 25,32,33, 38	13				
J8	BD04		38	4,5,18, 25,32,33, 37	36				

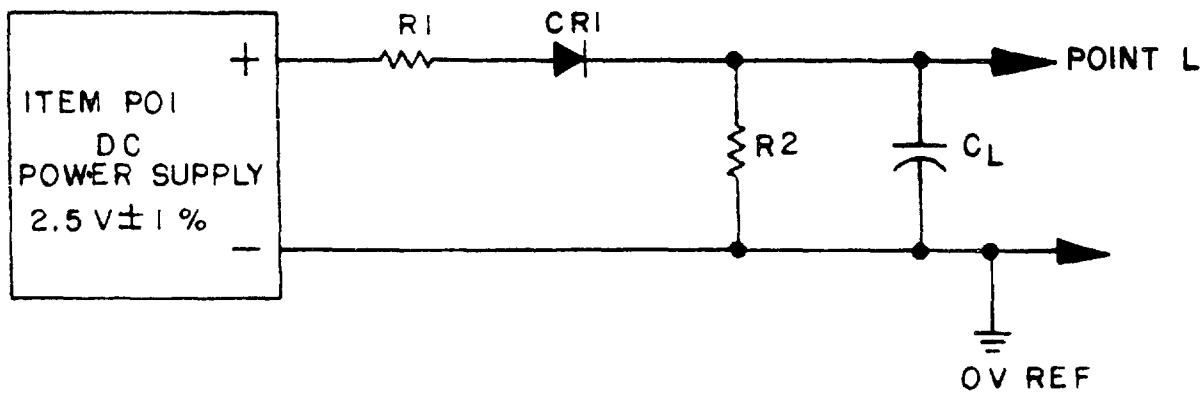
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NOTES.
1. R1 = 1.2 kΩ ±5%, 1/8 W.
2. R2 = R3 = 51Ω ±5%, 1/4 W.
3. S1 = SPST switch.
4. Close switch S1 for a minimum of 500 ns, then open and make measurements.
5. After V_{CC} has been applied do not remove it until test is complete.

* FIGURE 13. Output voltage (low-level and high-level) test circuit for table IX.

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NOTES:
1. CR1 = 1N3600, 1N4150 or equivalent.
2. R1 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V for BD01-BD04 outputs.
3. R1 = Select value for a current flow of 3.4 mA ±5% out of point L with point L held at 0.4 V for RD01-RD04 outputs.
4. R2 = 1.21 kΩ ±1%, 1/8 W for BD01-BD04 outputs.
   R2 = 2.49 kΩ ±1%, 1/8 W for RD01-RD04 outputs.
5. C_L < 10 pF (including probe and fixture capacitance) for test conditions S1-S16.
6. C_L = 50 ±5 pF (including probe and fixture capacitance) for test conditions Q1-Q8, Q17-Q24, R1-R8, T1-T4, T9-T12
7. C_L = 150 ±5 pF (including probe and fixture capacitance) for test conditions Q9-Q16, Q25-Q32, R9-R16, T5-T8, T13-T16.
8. Capacitor C_L is not required for dc electrical test conditions H1-H8 and J1-J8.
9. Lead length from load circuit to module output pin under test is to be less than two inches (50.8 mm)

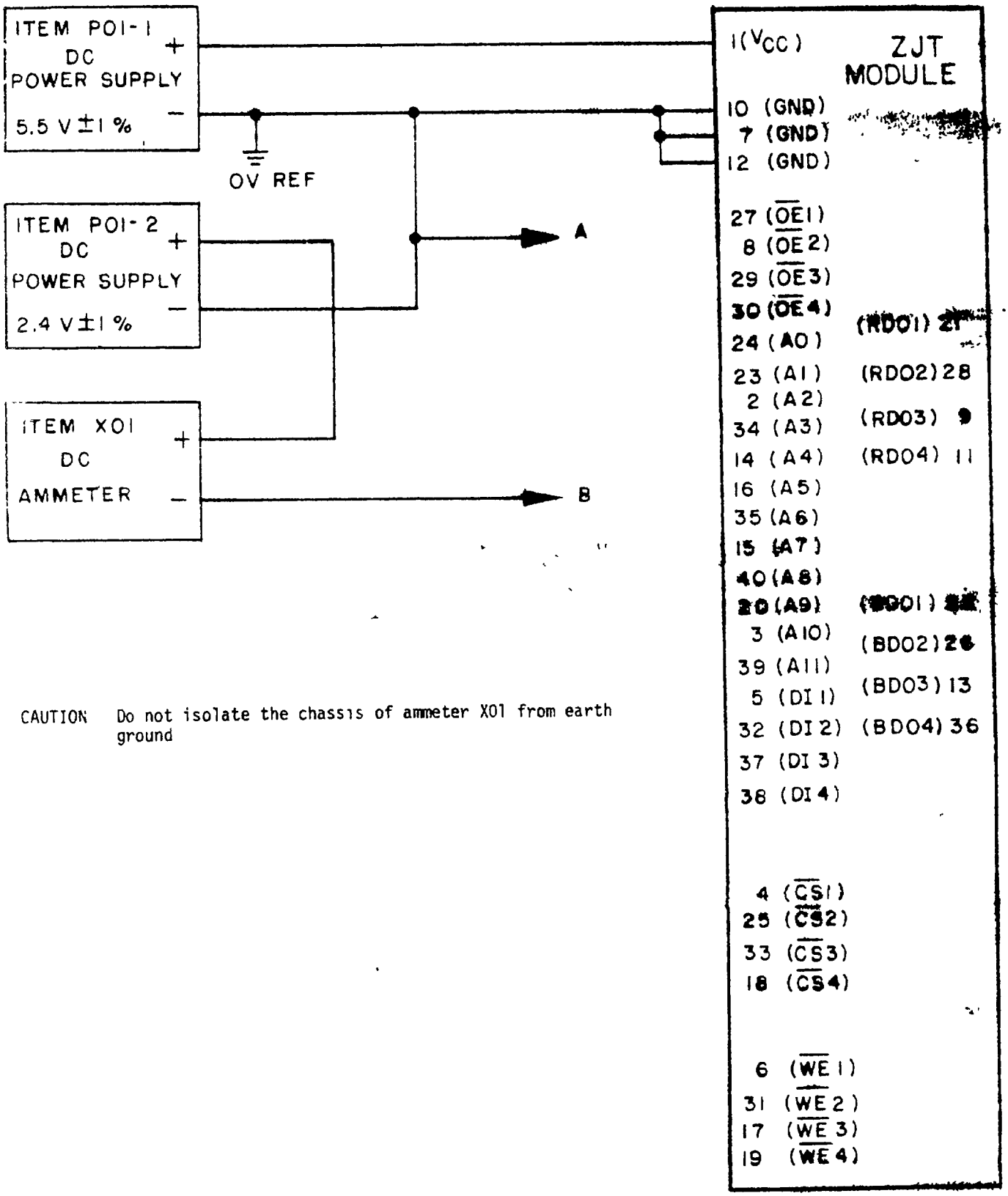
FIGURE 14. ZJT logic load circuit.

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TABLE X. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	$\overline{OE}1$ Input current, high level	8,29,30	27	0	40	µA	15
K2	$\overline{OE}2$	27,29,30	8				
K3	$\overline{OE}3$	8,27,30	29				
K4	$\overline{OE}4$	8,27,29	30				

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CAUTION Do not isolate the chassis of ammeter XO1 from earth ground

FIGURE 15. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	$\overline{CS1}$	Input current, low level	2,5,6,8,14,15,16,17,3,18,19,20, 23,24,25,27,29,30,31,32,33,34, 35,37,38,40,39	4	-10	1	µA	16
L2	$\overline{CS2}$		2,4,5,6,8,14,15,16,17,3,18,19, 20,23,24,27,29,30,31,32,33,34, 35,37,38,40,39	25				
L3	$\overline{CS3}$		2,4,5,6,8,14,15,16,17,3,18,19, 20,23,24,25,27,29,30,31,32,34, 35,37,38,40,39	33				
L4	$\overline{CS4}$		2,4,5,6,8,14,15,16,17,3,19,20, 23,24,25,27,29,30,31,32,33,34, 35,37,38,40,39	18				
L5	A0		2,4,5,6,8,14,15,16,17,3,18,19, 20,23,25,27,29,30,31,32,33,34, 35,37,38,40,39	24				
L6	A1		2,4,5,6,8,14,15,16,17,3,18,19, 20,24,25,27,29,30,31,32,33,34, 35,37,38,40,39	23				
L7	A2		4,5,6,8,14,15,16,17,3,18,19, 20,23,24,25,27,29,30,31,32, 33,34,35,37,38,40,39	2				
L8	A3		2,4,5,6,8,14,15,16,17,3,18,19, 20,23,24,25,27,29,30,31,32,33, 35,37,38,40,39	34				
L9	A4		2,4,5,6,8,15,16,17,3,18,19,20, 23,24,25,27,29,30,31,32,33,34, 35,37,38,40,39	14				

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TABLE XI. Input current (low-level) test - Continued.

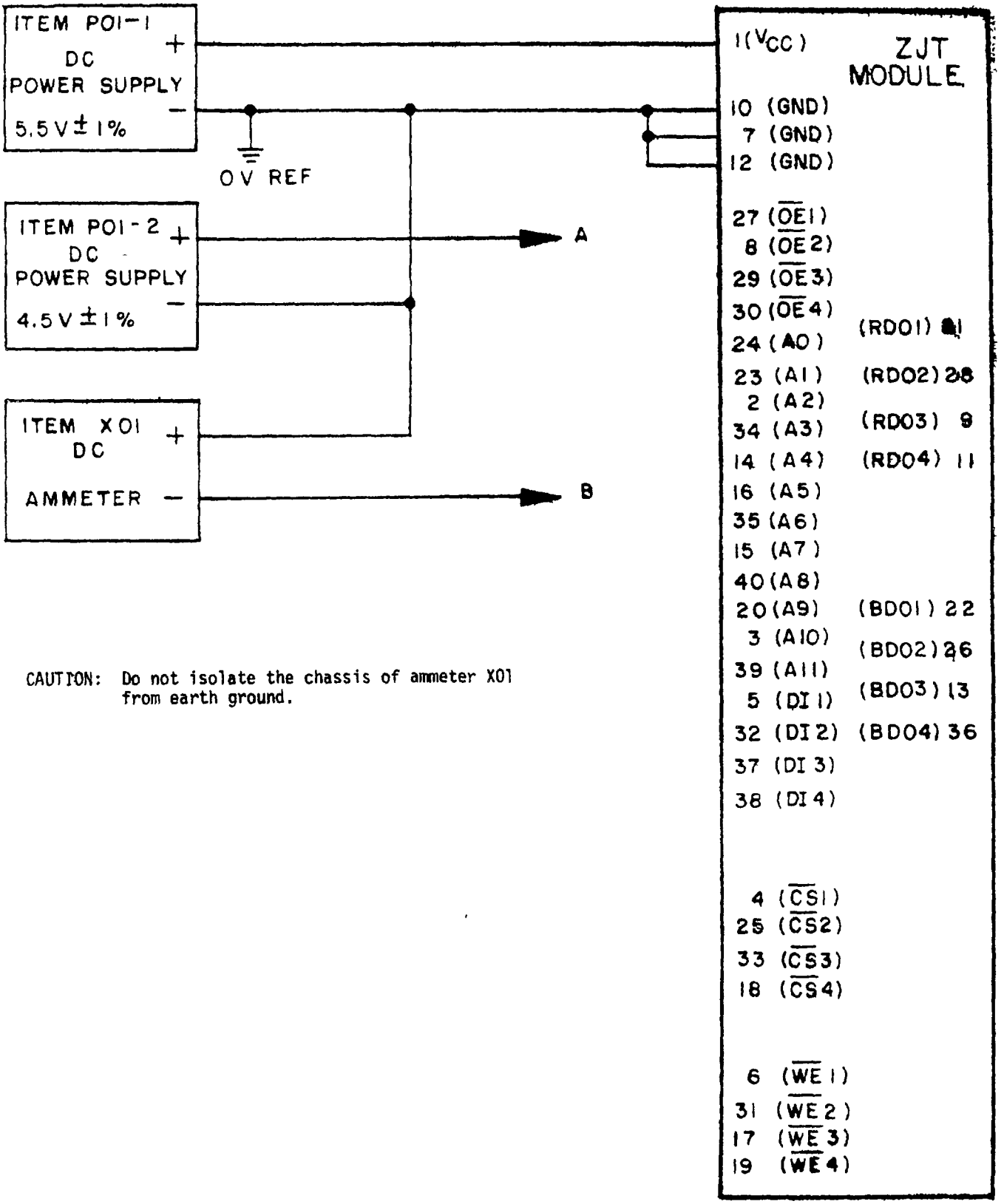
Test condition	Parameter		Connect		Initial limits 25 ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L10	A5	Input current, low level	2,4,5,6,8,14,15,16,17,3,18,19, 20,23,24,25,27,29,30,31,32,33, 34,35,37,38,40,39	16	-10	1	µA	16
L11	A6		2,4,5,6,8,14,15,16,17,3,18,19, 20,23,24,25,27,29,30,31,32,33, 34,37,38,40,39	35				
L12	A7		2,4,5,6,8,14,16,17,3,18,19,20, 23,24,25,27,29,30,31,32,33,34, 35,37,38,40,39	15				
L13	A8		2,4,5,6,8,14,15,16,17,3,18,19, 20,23,24,25,27,29,30,31,32,33, 34,35,37,38,39	40				
L14	A9		2,4,5,6,8,14,15,16,17,3,18,19, 23,24,27,29,30,31,32,33,34,35, 37,38,40,39,25	20				
L15	A10		2,4,5,6,8,14,15,16,17,18,19,20, 23,24,25,27,29,30,31,32,33,34, 35,37,38,40,39	3				
L16	A11		2,4,5,6,8,14,15,16,17,18,19,20, 3,23,24,25,31,32,33,34,35,37, 38,40,29,30,27	39				
L17	$\overline{WE1}$		2,4,5,8,14,15,16,17,3,18,19,20, 23,24,25,27,29,30,31,32,33,34, 35,37,38,40,39	6				
L18	$\overline{WE2}$		2,4,5,6,8,14,15,16,17,3,18,19, 20,23,24,27,29,30,25,32,33,34, 35,37,38,40,39	31				

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TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
L19	$\overline{WE3}$	Input current, low level	2,4,5,6,8,14,15,16,3,18,19, 20,23,24,25,27,29,30,31,32,33, 34,35,37,38,40,39	17	-10	1	µA	16
L20	$\overline{WE4}$		2,4,5,6,8,14,15,16,17,3,18, 20,23,24,25,27,29,30,31,32, 33,34,35,37,38,40,39	19				
L21	DI1		2,4,6,8,14,15,16,17,3,18,19, 20,23,24,25,27,29,30,31,32, 33,34,35,37,38,40,39	5				
L22	DI2		2,4,5,6,8,14,15,16,17,3,18, 19,20,23,24,25,27,29,30,31, 33,34,35,37,38,40,39	32				
L23	DI3		2,4,5,6,8,14,15,16,17,3,18, 19,20,23,24,25,27,29,30,31, 32,33,34,35,38,40,39	37				
L24	DI4		2,4,5,6,8,14,15,16,17,3,18, 19,20,23,24,25,27,29,30,31, 32,33,34,35,37,40,39	38				
L25	$\overline{OE1}$		2,4,5,6,8,14,15,16,17,3,18, 19,20,23,24,25,29,30,31,32, 33,34,35,37,38,39,40	27	-0.1	-1.73	mA	
L26	$\overline{OE2}$		2,3,4,5,6,14,15,16,17,18,19, 20,23,24,25,27,29,30,31,32, 33,34,35,37,38,39,40	8				
L27	$\overline{OE3}$		2,3,4,5,6,8,14,15,16,17,18,19, 20,23,24,25,27,30,31,32,33,34, 35,37,38,39,40	29				
L28	$\overline{OE4}$		2,3,4,5,6,8,14,15,16,17,18,19, 20,23,24,25,27,29,31,32,33,34, 35,37,38,39,40	30				

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CAUTION: Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 16. Input current (low-level) test circuit for table XI.

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* TABLE XII. Address input threshold check (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Address input threshold check, low level	6,17,19,31	2,14,15, 16,20,23, 24,34,35, 40,4,25, 18,3,39, 33	21	0	400	mV	17
M2				28				
M3				9				
M4				11				
M5	Address input threshold check, high level	2,6,14,15, 16,17,19, 20,23,24, 31,34,35, 40,3,39	4,18,25, 33	21	2.4	4.5	V	
M6				28				
M7				9				
M8				11				

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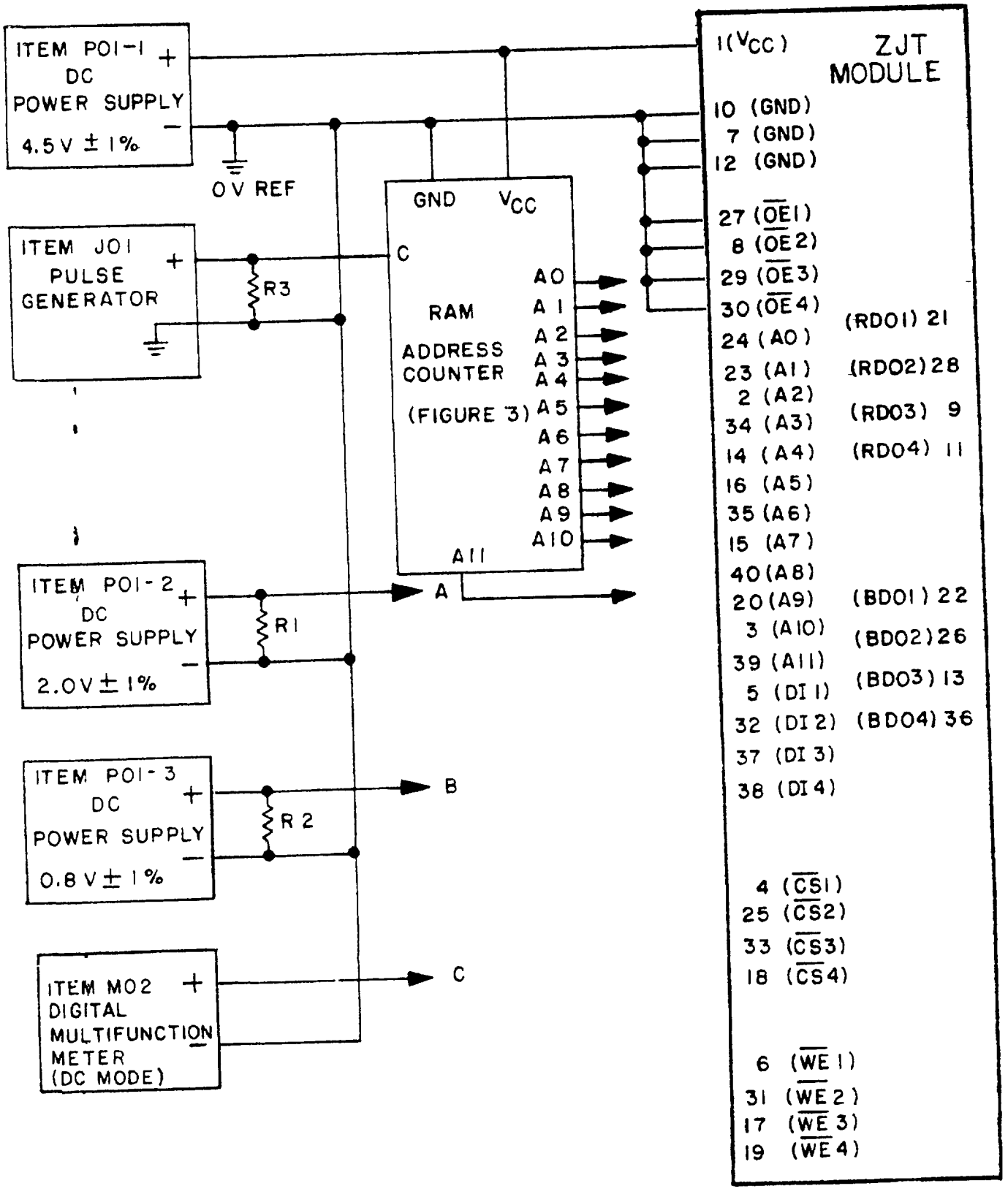


FIGURE 17. Address input threshold check (low-level and high-level) test circuit for table XII.

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NOTES:
1. R1-R3 = 51Ω ±5%, 1 W.
2. Load all outputs with the load as specified on figure 14.
3. Item J01 parameters: (see figure 21 for parameter definition)
   a. PA = 4.0 ±0.1 V.
   b. PWA = 1000 ±50 ns.
   c. PRP = 2000 ns ±10%.
   d. $t_{TLH} = t_{THL}$ = 25 ±20 ns.
4. Test sequence for test conditions M1-M4:
   a. Connect RAM address counter to address inputs.
   b. Connect pins 5, 32, 37, and 38 to A. Connect pins 4, 6, 17, 18, 19, 25, 31 and 33 to B.
   c. Initiate RAM address counter for five seconds. This loads all memory locations with a high level.
   d. Disconnect pins 6, 17, 19 and 31 from B and connect to A.
   e. Disconnect the RAM address counter. Connect pins 2, 14, 15, 16, 20, 3, 39, 23, 24, 34, 35 and 40 to B. Connect pins 5, 32, 37 and 38 to B.
   f. Verify the outputs at location zero are high.
   g. Disconnect pins 6, 17, 19 and 31 from A and connect to B for a minimum of 500 ns, then connect to point A. Verify that this loads memory location zero with a low level and all other locations with a high level.
   h. Without disconnecting pins 6, 17, 19 and 31 from point A and without interrupting power perform tests as specified in test conditions M1-M4.
5. Test sequence for test conditions M5-M8.

   a. Connect RAM address counter to address inputs.
   b. Connect pins 4, 5, 6, 17, 18, 19, 25, 31, 32, 33, 37, and 38 to B.
   c. Initiate RAM address counter for five seconds. This loads all memory locations with a low level.
   d Disconnect pins 6, 17, 19 and 31 from B and connect to A.
   e. Disconnect the RAM address counter. Connect pins 2, 14, 15, 16, 20, 3, 39, 23, 24, 34, 35 and 40 to A. Connect pins 5, 32, 37 and 38 to A.
   f. Verify the outputs at locations 4095 are low.
   g. Disconnect pins 6, 17, 19 and 31 from A and connect to B for a minimum of 500 ns, then connect to point A Verify that this loads memory location 4095 with a high level and all other locations with a low level.
   h. Without disconnecting pins 6, 17, 19 and 31 from point A and without interrupting power perform tests as specified in test conditions M5-M8.

FIGURE 17. Address input threshold check (low-level and high-level) test circuit for table XII - Continued.

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* TABLE XIII. Ground pin continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Ground pin continuity	10	7	0	1	Ω	18
N2		10	12	0	1		
N3		7	12	0	1		
N4		10	7	0.0	0.1		
N5		10	12	0.0	0.1		
N6		7	12	0.0	0.1		

TABLE XIV. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Power supply current	4,25,33,18	Open	50	774	mA	19
P2		Open	4,25,33,18	20	174		
P3		4	18,25,33	50	324		
P4		25	4,18,33				
P5		33	4,18,25				
P6		18	4,25,33				

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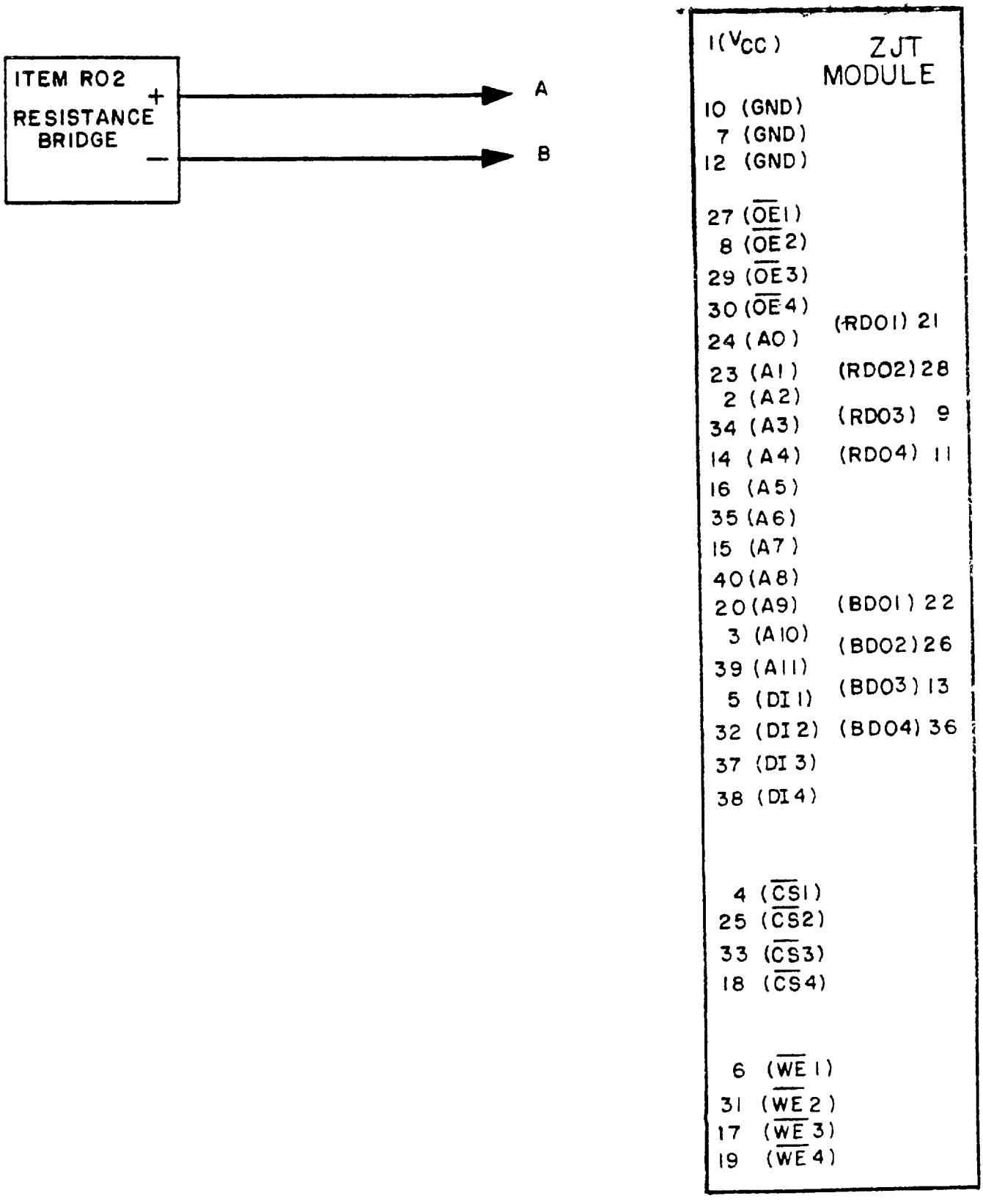


FIGURE 18. Ground pin continuity test circuit for table XIII.

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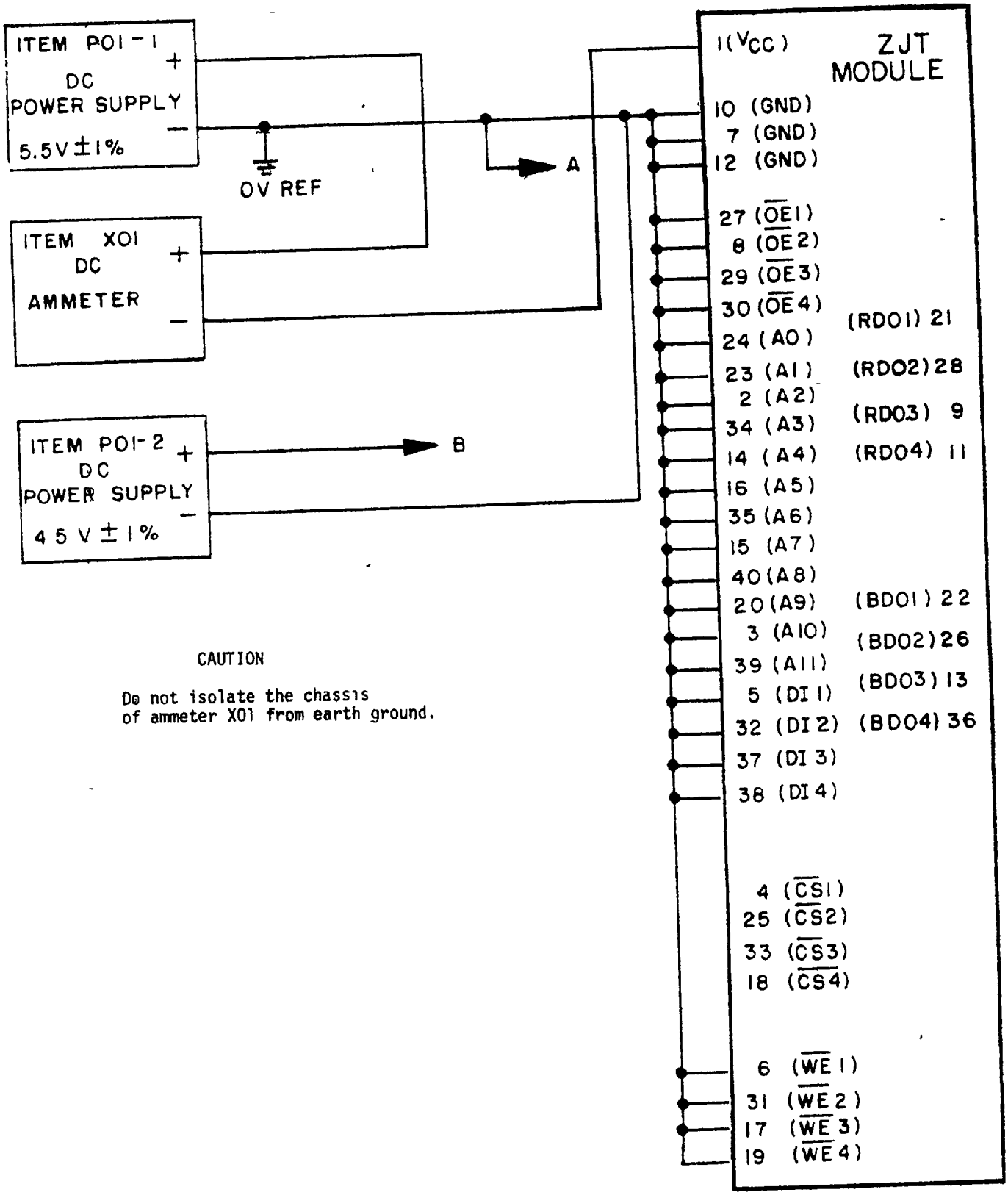


FIGURE 19. Power supply current test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

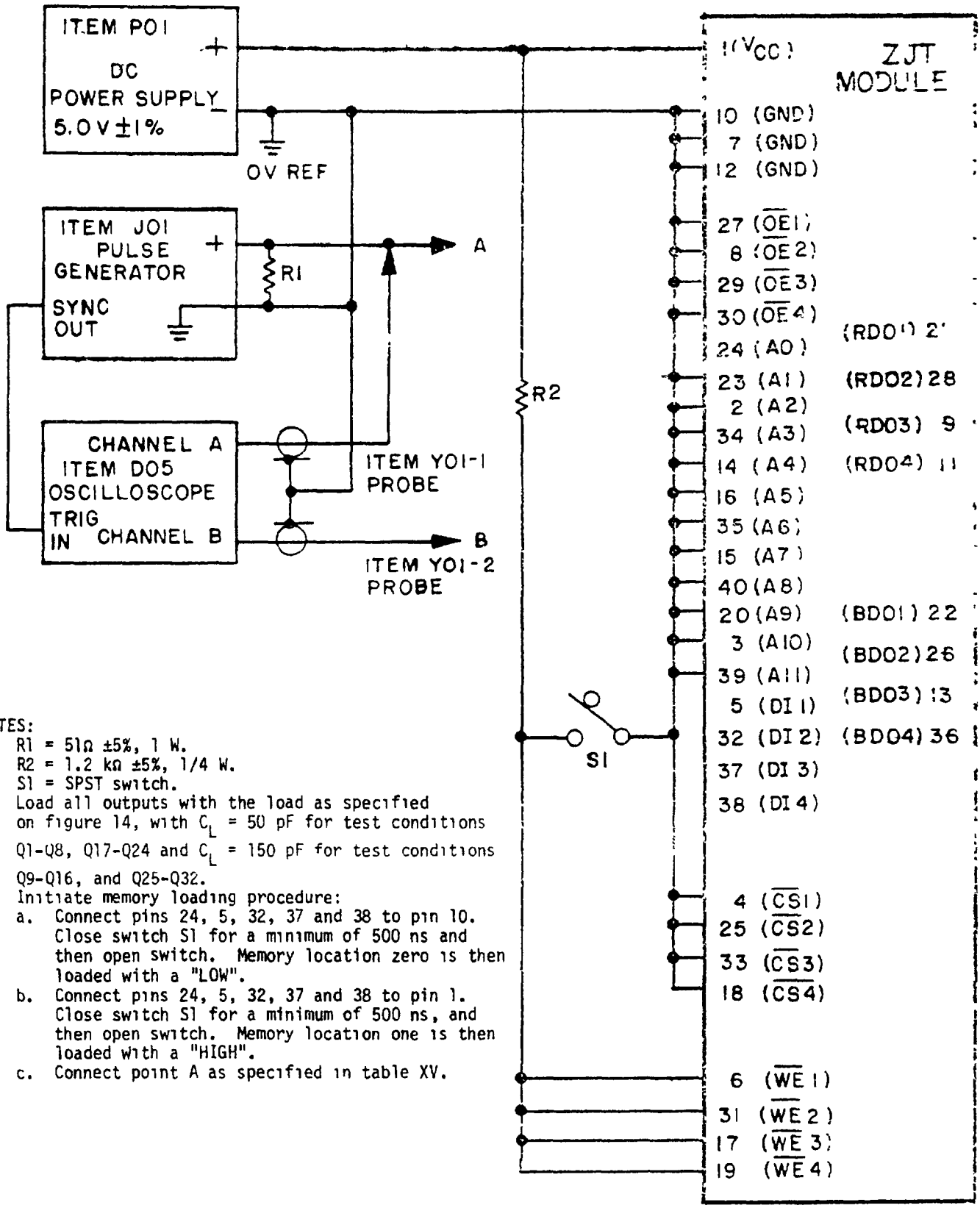
Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Q1	RD01	Transition time (t_{TLH}), C_L = 50 pF	24	21	1	10	ns	20,21
Q2	RD02			28				
Q3	RD03			9				
Q4	RD04			11				
Q5	BD01			22				
Q6	BD02			26				
Q7	BD03			13				
Q8	BD04			36				
Q9	RD01	Transition time (t_{TLH}), C_L = 150 pF		21		13		
Q10	RD02			28				
Q11	RD03			9				
Q12	RD04			11				
Q13	BD01			22				
Q14	BD02			26				
Q15	BD03			13				
Q16	BD04			36				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Q17	RD01	Transition time (t_{THL}), C_L = 50 pF	24	21	1	7	ns	20,21
Q18	RD02			28				
Q19	RD03			9				
Q20	RD04			11				
Q21	BD01			22				
Q22	BD02			26				
Q23	BD03			13				
Q24	BD04			36				
Q25	RD01	Transition time (t_{THL}), C_L = 150 pF		21		10		
Q26	RD02			28				
Q27	RD03			9				
Q28	RD04			11				
Q29	BD01			22				
Q30	BD02			26				
Q31	BD03			13				
Q32	BD04			36				

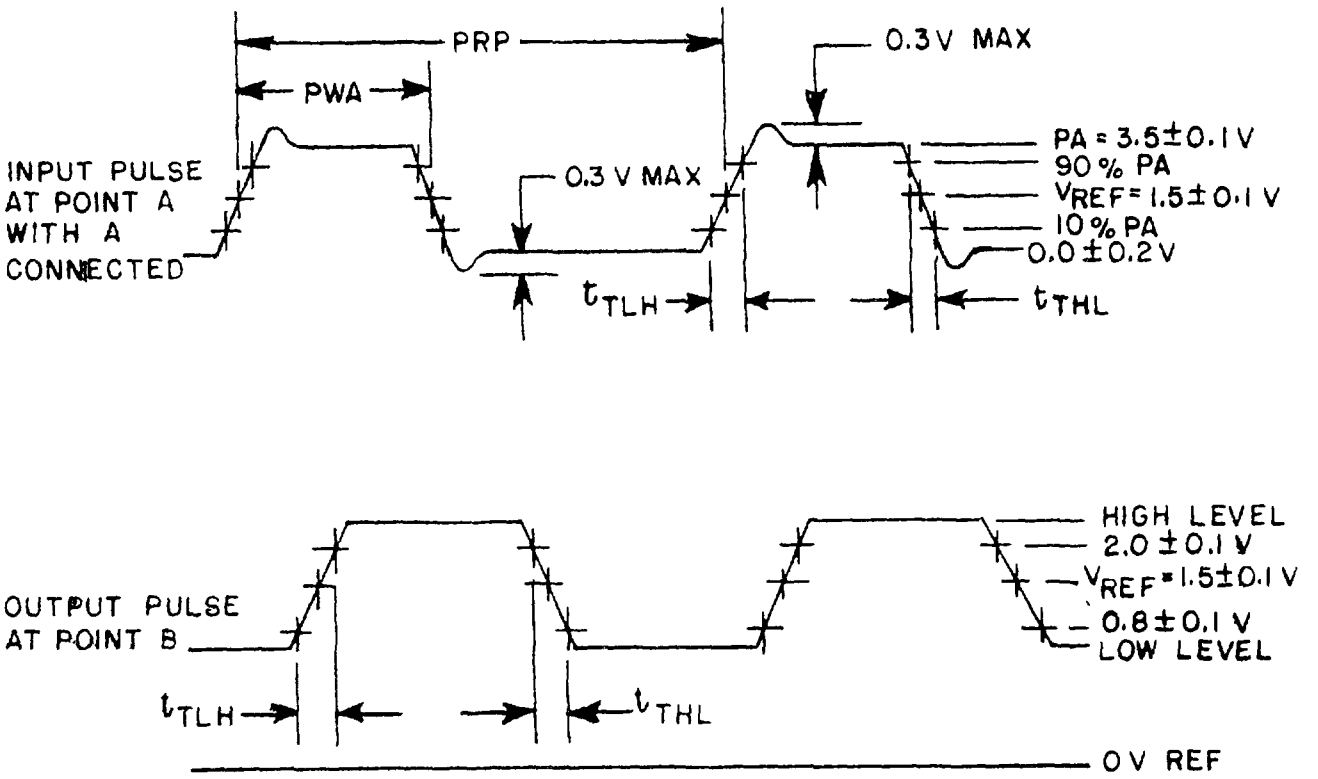
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:33Z
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1.2 kΩ ±5%, 1/4 W.
3. S1 = SPST switch.
4. Load all outputs with the load as specified on figure 14, with C_L = 50 pF for test conditions Q1-Q8, Q17-Q24 and C_L = 150 pF for test conditions Q9-Q16, and Q25-Q32.
5. Initiate memory loading procedure:
   a. Connect pins 24, 5, 32, 37 and 38 to pin 10. Close switch S1 for a minimum of 500 ns and then open switch. Memory location zero is then loaded with a "LOW".
   b. Connect pins 24, 5, 32, 37 and 38 to pin 1. Close switch S1 for a minimum of 500 ns, and then open switch. Memory location one is then loaded with a "HIGH".
   c. Connect point A as specified in table XV.

FIGURE 20. Transition time (t_{TLH} and t_{THL}) test circuit for table XV.

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NOTES:
1. Input pulse parameters.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PWA = 1000 ±100 ns.
   PRP = 2000 ±200 ns.
2. Before making the measurements verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.40 V.
3. All input voltages are referenced to ground

FIGURE 21. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XV.

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TABLE XVI. Dynamic test.

Test condition	Parameter	Initial limits 25° ±5°C ambient	Test figure
R1	Inter-address write recovery-one, inter-address write recovery-zero pattern (C_L = 50 pF, V_{CC} = 4.5 V).	Modules shall conform to figure 24 requirements	22,23,24
R2	Inter-address write recovery-one, inter-address write recovery-zero pattern (C_L = 50 pF, V_{CC} = 5.5 V).	Modules shall conform to figure 24 requirements	22,23,24
R3	Marching write recovery-one, marching write recovery-zero, and circuit-select/deselect and chip independence pattern (C_L = 50 pF, V_{CC} = 4.5 V).	Modules shall conform to figure 25 requirements	22,23,25
R4	Marching write recovery-one, marching write recovery-zero, and circuit-select/deselect and chip independence pattern (C_L = 50 pF, V_{CC} = 5.5 V).	Modules shall conform to figure 25 requirements	22,23,25
R5	Static hold test pattern (C_L = 50 pF, V_{CC} = 4.5 V).	Modules shall conform to figure 26 requirements	22,23,26
R6	Static hold test pattern (C_L = 50 pF, V_{CC} = 5.5 V).	Modules shall conform to figure 26 requirements	22,23,26
R7	Minimum deselect time pattern (C_L = 50 pF, V_{CC} = 4.5 V).	Modules shall conform to figure 27 requirements	22,23,27
R8	Minimum deselect time pattern (C_L = 50 pF, V_{CC} = 5.5 V).	Modules shall conform to figure 27 requirements	22,23,27

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TABLE XVI. Dynamic test - Continued.

Test condition	Parameter	Initial limits 25° ±5°C ambient	Test figure
R9	Inter-address write recovery-one, inter-address write recovery-zero pattern (C_L = 150 pF, V_{CC} = 4.5 V).	Modules shall conform to figure 24 requirements	22,23,24
R10	Inter-address write recovery-one, inter-address write recovery-zero pattern (C_L = 150 pF, V_{CC} = 5.5 V).	Modules shall conform to figure 24 requirements	22,23,24
R11	Marching write recovery-one, marching write recovery-zero, and circuit-select/deselect and chip independence pattern (C_L = 150 pF, V_{CC} = 4.5 V).	Modules shall conform to figure 25 requirements	22,23,25
R12	Marching write recovery-one, marching write recovery-zero, and circuit-select/deselect and chip independence pattern (C_L = 150 pF, V_{CC} = 5.5 V).	Modules shall conform to figure 25 requirements	22,23,25
R13	Static hold test pattern (C_L = 150 pF, V_{CC} = 4.5 V).	Modules shall conform to figure 26 requirements	22,23,26
R14	Static hold test pattern (C_L = 150 pF, V_{CC} = 5.5 V).	Modules shall conform to figure 26 requirements	22,23,26
R15	Minimum deselect time pattern (C_L = 150 pF, V_{CC} = 4.5 V).	Modules shall conform to figure 27 requirements	22,23,27
R16	Minimum deselect time pattern (C_L = 150 pF, V_{CC} = 5.5 V).	Modules shall conform to figure 27 requirements	22,23,27

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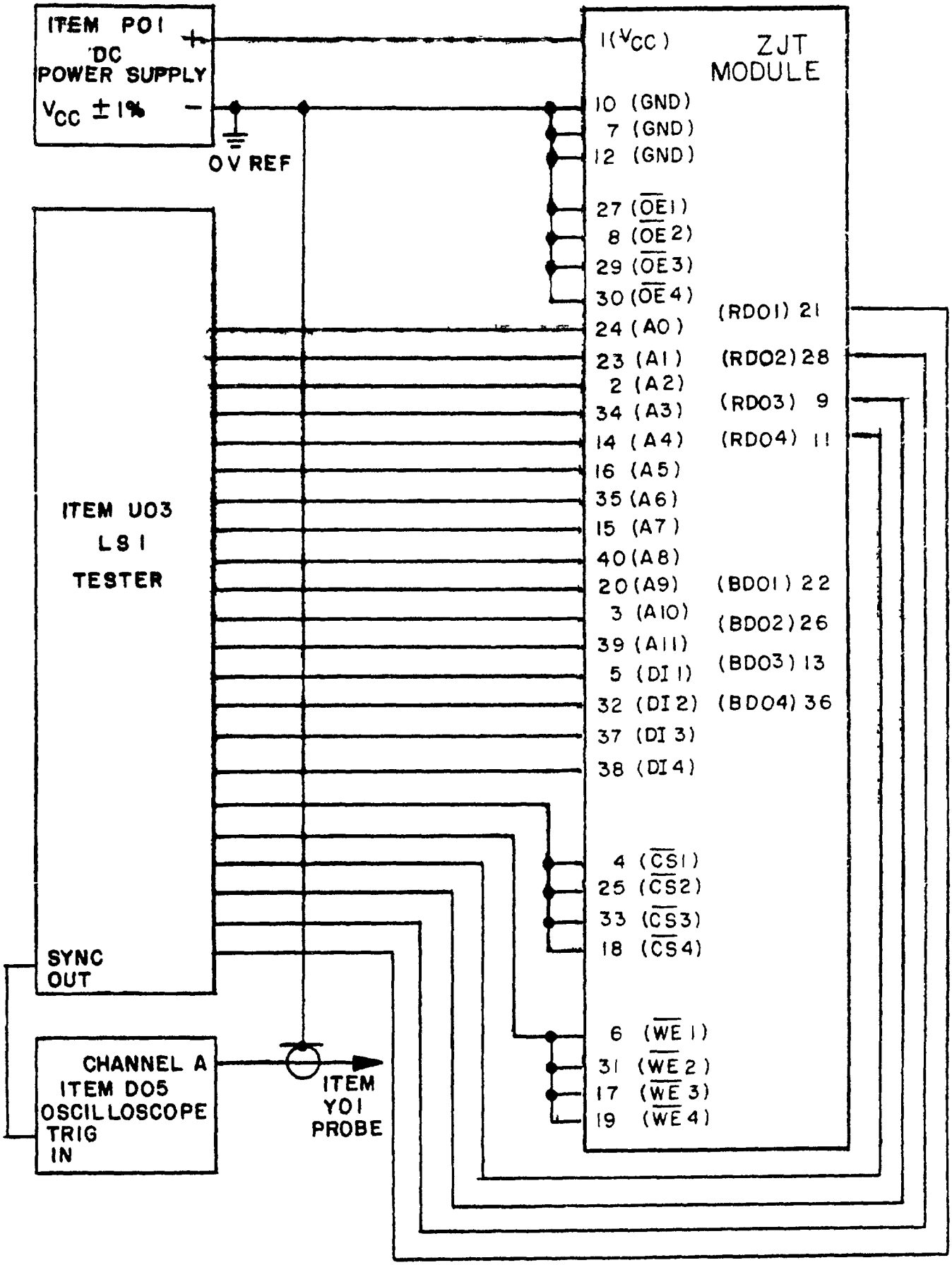


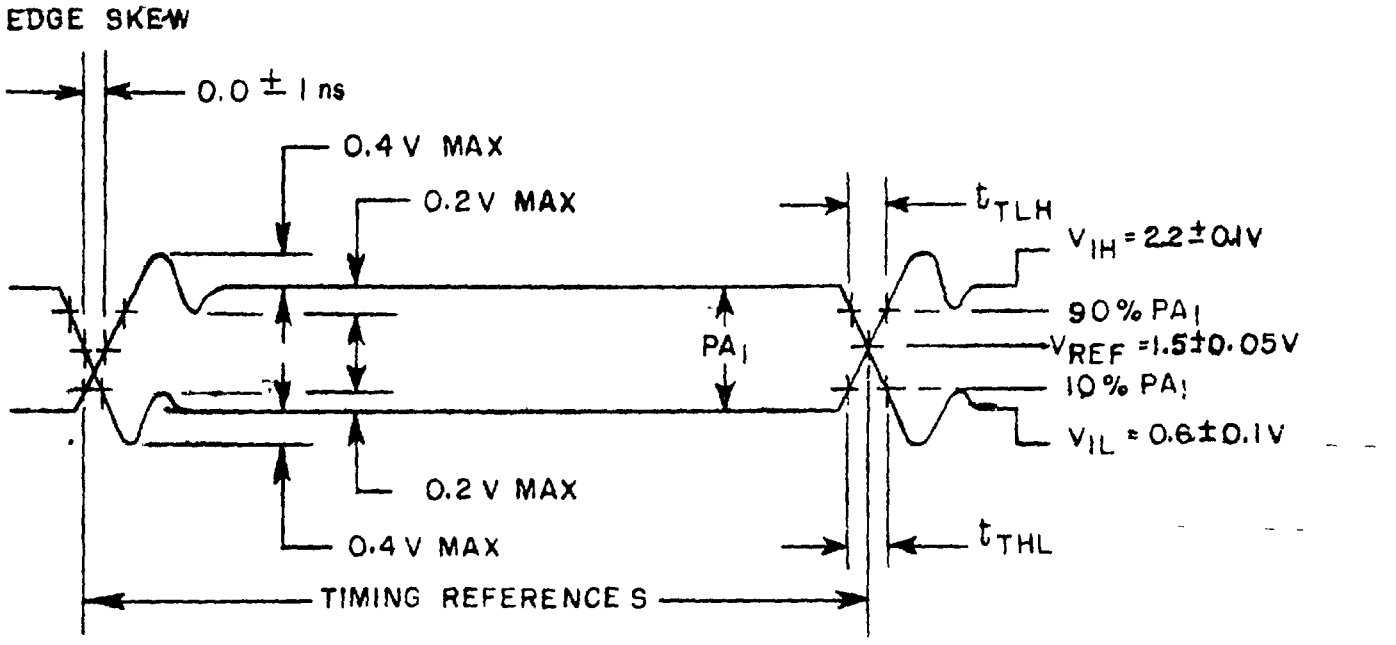
FIGURE 22. Dynamic test circuit for table XVI.

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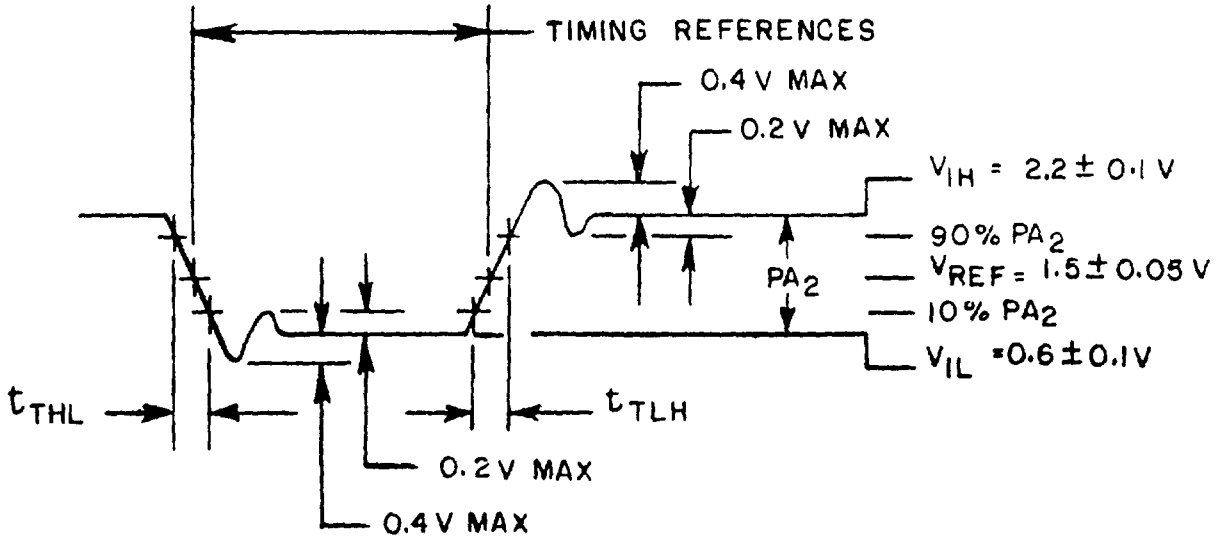
NOTES:
1. V_{CC} = 4.5 V ±1% for test conditions R1, R3, R5, R7, R9, R11, R13, and R15.
2. V_{CC} = 5.5 V ±1% for test conditions R2, R4, R6, R8, R10, R12, R14, and R16.
3. Load output pins 9, 11, 21, and 28 with load as specified on figure 14 with C_L = 50 pF for test conditions R1-R8.
4. Load output pins 9, 11, 21, and 28 with load as specified on figure 14 with C_L = 150 pF for test conditions R9-R16.
5. See Drawing No. 3202946G1 for item U03/module interface definition.

FIGURE 22. Dynamic test circuit for table XVI - Continued.

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BIDIRECTIONAL INPUT WAVEFORMS



UNIDIRECTIONAL INPUT WAVEFORMS

NOTES: 1. Input pulse parameters.
a. $t_{THL} = t_{TLH} = 10 \pm 4$ ns.
b. All timing references are made at the V_{REF} level.

FIGURE 23. Dynamic test circuit waveforms and waveform parameters for table XVI.

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2. Waveform parameters:

Parameter	Test conditions R1, R2, R9, R10	Test conditions R3, R4, R11, R12	Test conditions R5, R6 R13, R14	Test conditions R7, R8, R15, R16	Unit
a. Address setup time (t_{AS})	-	20 +0 -3	20 +0 -3	20 +0 -3	ns
b. Data setup time (t_{DS})	40 +0 -4	40 +0 -4	40 +0 -4	40 +0 -4	ns
c. Circuit select setup time (t_{SS})	-	70 +0 -5	70 +0 -5	-	ns
d Deselect setup time ($t_{\overline{SS}}$)	-	10 +0 -2	-	-	ns
e Address hold time (t_{AH})	15 +0 -3	-	-	-	ns
f. Data hold time (t_{DH})	10 +0 -3	-	10 +0 -3	10 +0 -3	ns
g. Circuit select hold time (t_{SH})	10 +0 -3	10 +0 -3	10 +0 -3	-	ns
h. Write pulse width (t_{WP})	50 +0 -5	50 +0 -5	50 +0 -5	50 +0 -5	ns
i Address access time (t_{AA})					
RAM output, C_L = 50 pF	90 +0 -5	90 +0 -5	90 +0 -5	105 +0 -5	ns
RAM output, C_L = 150 pF	100 +0 -5	100 +0 -5	100 +0 -5	115 +0 -5	ns
Buffer output, C_L = 50 pF	110 +0 -5	110 +0 -5	110 +0 -5	125 +0 -5	ns
Buffer output, C_L = 150 pF	120 +0 -5	120 +0 -5	120 +0 -5	135 +0 -5	ns
j. Circuit select access time (t_{SA}) (Chip deselected for greater than 55 ns prior to selection)					
RAM output, C_L = 50 pF	-	90 +0 -5	90 +0 -5	-	ns
RAM output, C_L = 150 pF	-	100 +0 -5	100 +0 -5	-	ns
Buffer output, C_L = 50 pF	-	110 +0 -5	110 +0 -5	-	ns
Buffer output, C_L = 150 pF	-	120 +0 -5	120 +0 -5	-	ns

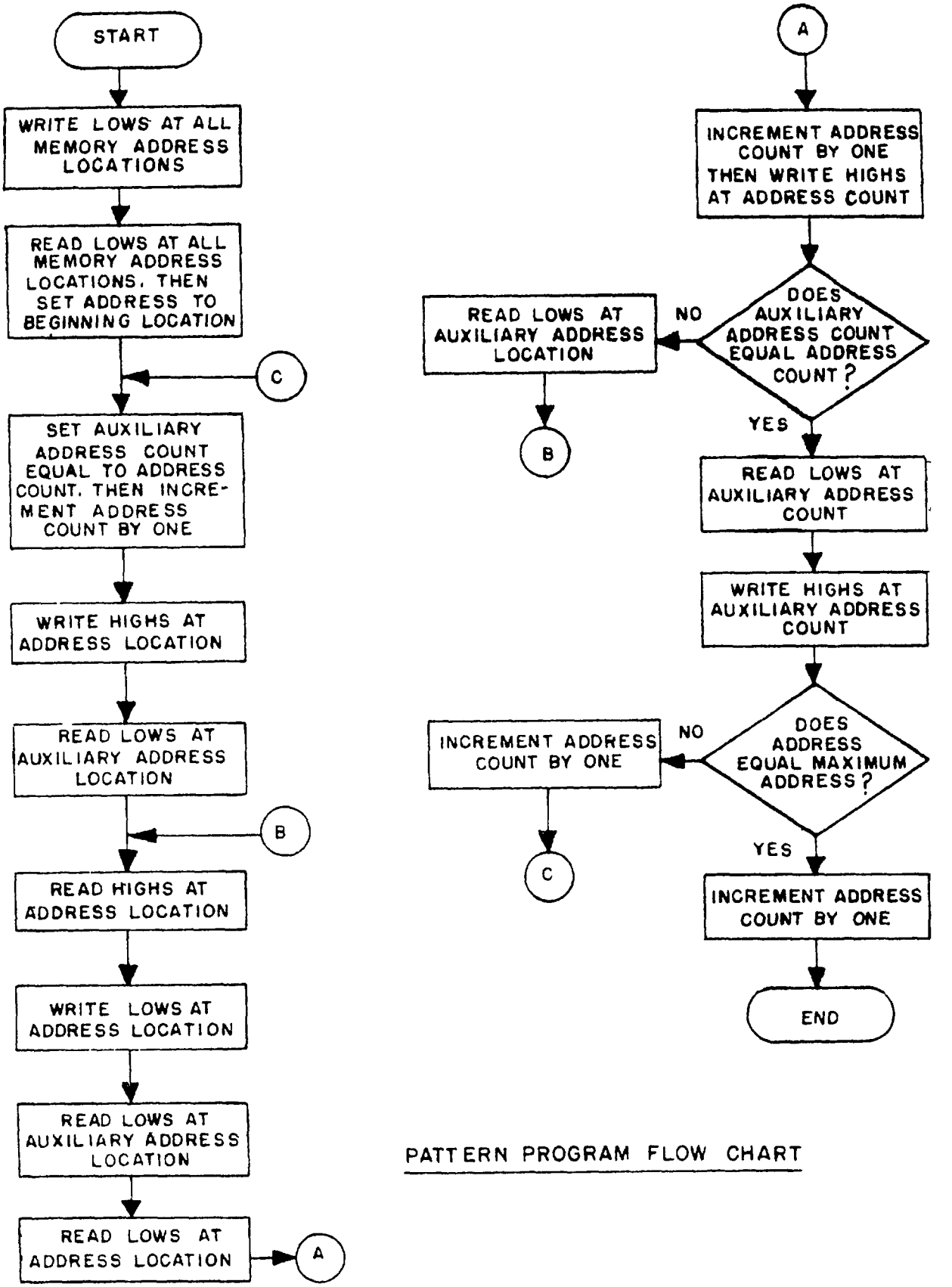
FIGURE 23. Dynamic test circuit waveforms and waveform parameters for table XVI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:33Z
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	Test conditions R1, R2, R9, R10	Test conditions R3, R4, R11, R12	Test conditions R5, R6, R13, R14	Test conditions R7, R8 R15, R16	Unit
k. Circuit select access time (t_{SA}) (Chip deselected for less than 55 ns prior to selection)					
RAM output, C_L = 50 pF	-	-	-	105 +0 -5	ns
RAM output, C_L = 150 pF	-	-	-	115 +0 -5	ns
Buffer output, C_L = 50 pF	-	-	-	125 +0 -5	ns
Buffer output, C_L = 150 pF	-	-	-	135 +0 -5	ns
l. Write recovery time (t_{WR})					
RAM output, C_L = 50 pF	-	50 +0 -3	-	-	ns
RAM output, C_L = 150 pF	-	60 +0 -3	-	-	ns
Buffer output, C_L = 50 pF	-	70 +0 -3	-	-	ns
Buffer output, C_L = 150 pF	-	80 +0 -5	-	-	ns
m. Minimum deselect time (t_S)	-	-	-	20 +0 -2	ns

FIGURE 23. Dynamic test circuit waveforms and waveform parameters for table XVI - Continued.

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PATTERN PROGRAM FLOW CHART

FIGURE 24 Inter-address write recovery-one, inter-address write recovery-zero pattern program flow chart and waveforms for table XVI.

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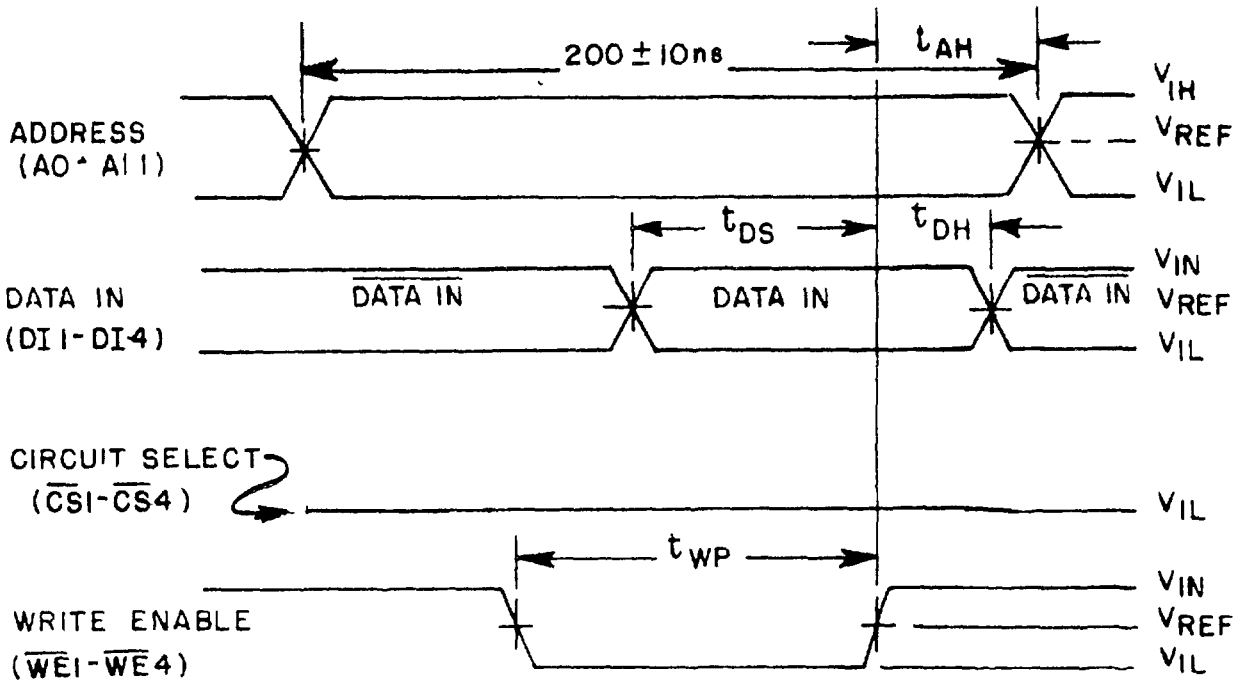
NOTES:

1. The objective of the inter-address write recovery-one, inter-address write recovery-zero test pattern is to test the recovery time from a write at one address location to read at a different address location. It is necessary to check all possible address transitions from a write cycle at one address to a read cycle at a different address. This pattern also verifies data setup time (t_{DS}), address hold time (t_{AH}), data hold time (t_{DH}), write pulse width (t_{WP}), address access time (t_{AA}).
2. A program that fulfills the inter-address write recovery-one test pattern requirements would

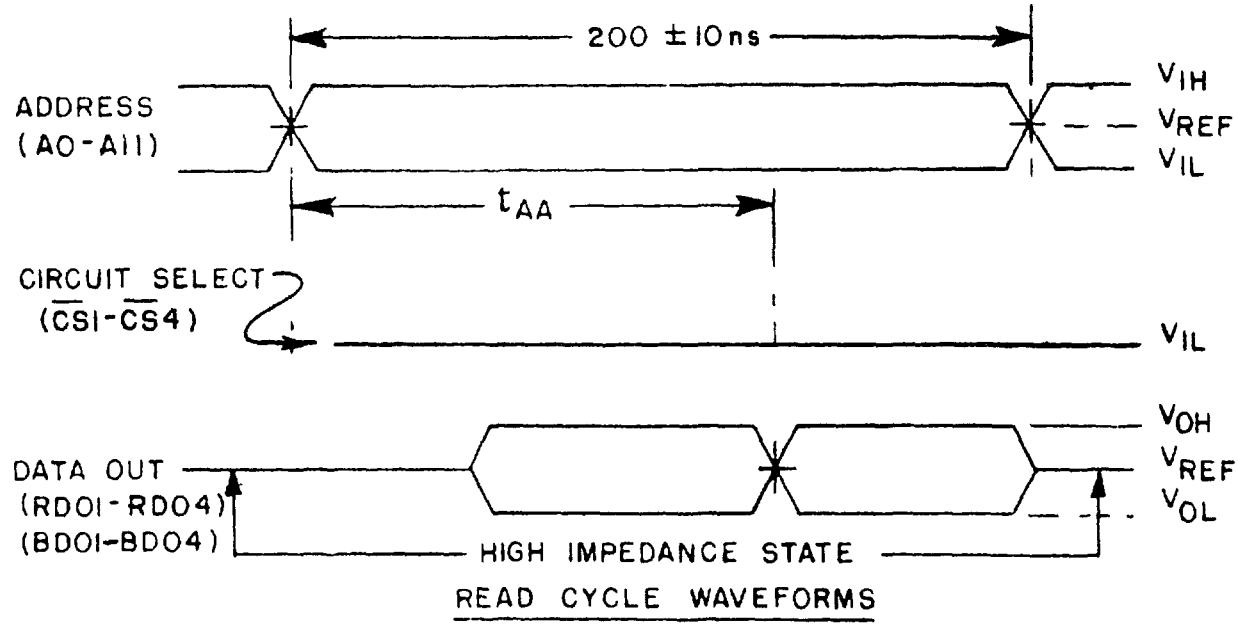
   a. Write "lows" at all memory address locations.
   b. Read "lows" at all memory address locations.
   c. Write "highs" at memory address locations one.
   d. Read "lows" at memory address location zero
   e. Read "highs" at memory address location one
   f. Write "lows" at memory address location one.
   g. Read "lows" at memory address location zero.
   h. Read "lows" at memory address location one.
   i. Continue the pattern until memory address location one has been checked with respect to all other memory address locations.
   j. Write "highs" at memory address location two
   k. Read "lows" at memory address location one.
   l Read "highs" at memory address location two.
   m. Write 'lows" at memory address location two.
   n. Read "lows" at memory address location one.
   o. Read "lows" at memory address location two.
   p. Write "highs" at memory address location zero.
   q Read "lows" at memory address location one.
   r. Read "highs" at memory address location zero.
   s. Write "lows" at memory address location zero.
   t. Read "lows" at memory address location one.
   u. Read "lows" at memory address location zero.
   v. Continue the pattern until all memory address locations have been checked with respect to all other memory address locations.
3. All address lines need not switch simultaneously each cycle.
4. A program that fulfills the inter-address write recovery-zero test pattern requirements would be as in note 2 above with all "highs" and "lows" interchanged.

FIGURE 24 Inter-address write recovery-one, inter-address write recovery-zero pattern program flow chart and waveforms for table XVI - Continued.

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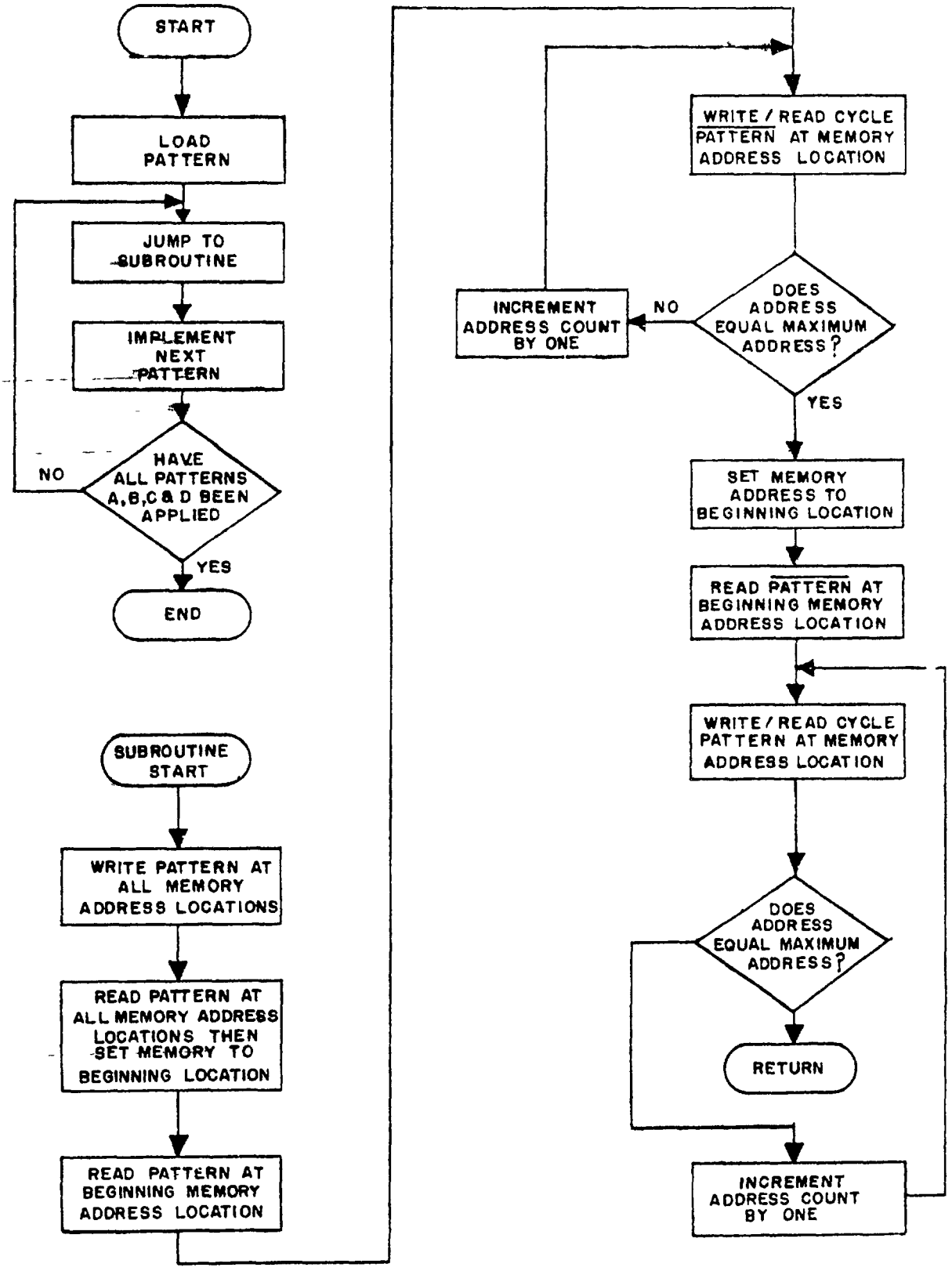
WRITE CYCLE WAVEFORMS



READ CYCLE WAVEFORMS

FIGURE 24. Inter-address write recovery-one, inter-address write recovery-zero pattern program flow chart and waveforms for table XVI - Continued.

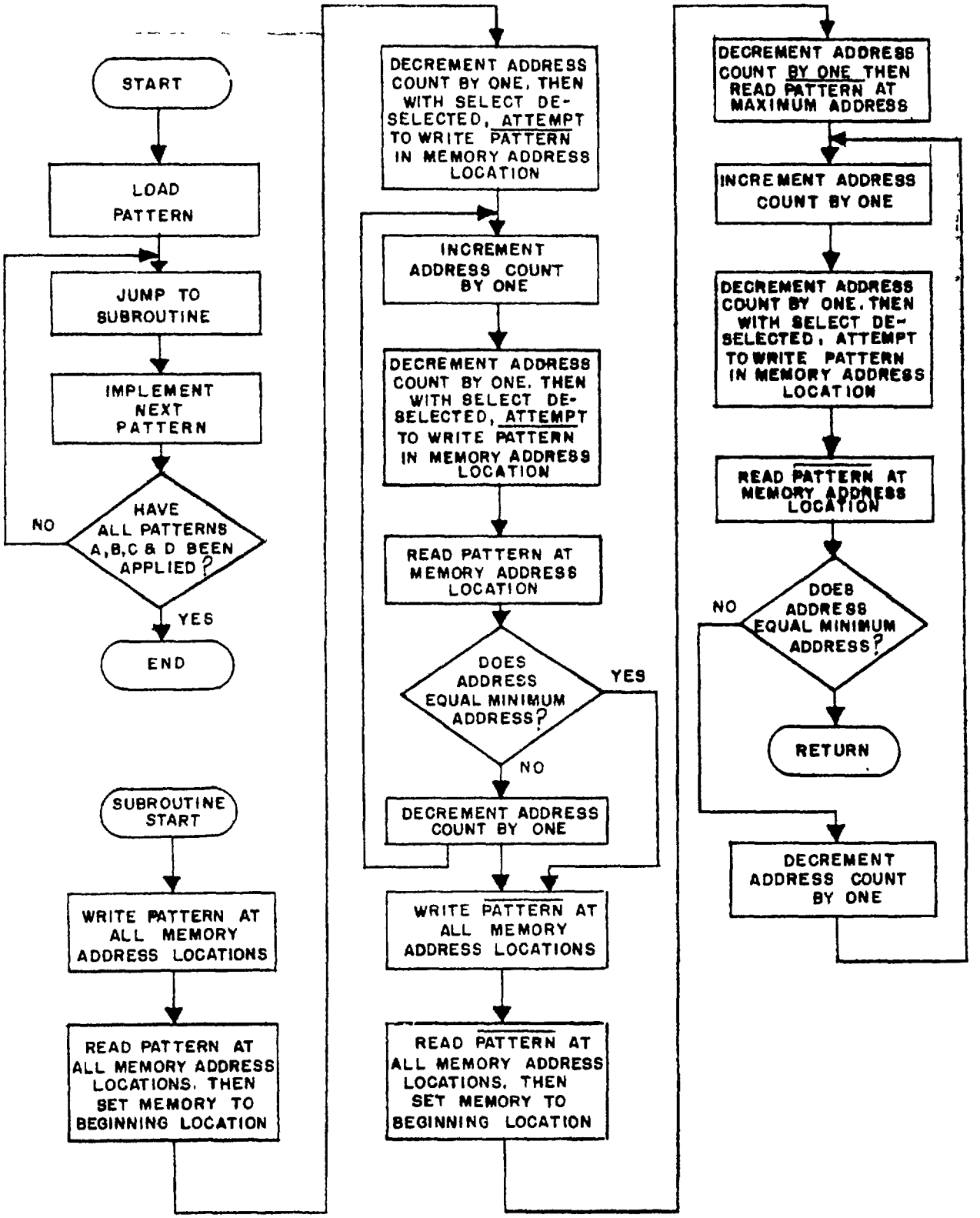
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:33Z
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MARCHING WRITE RECOVERY-ONE, MARCHING WRITE RECOVERY-ZERO PATTERN PROGRAM FLOW CHART.

FIGURE 25. Marching write recovery-one, marching write recovery-zero, and circuit select/deselect and chip independence pattern program flow charts and waveforms for table XVI.

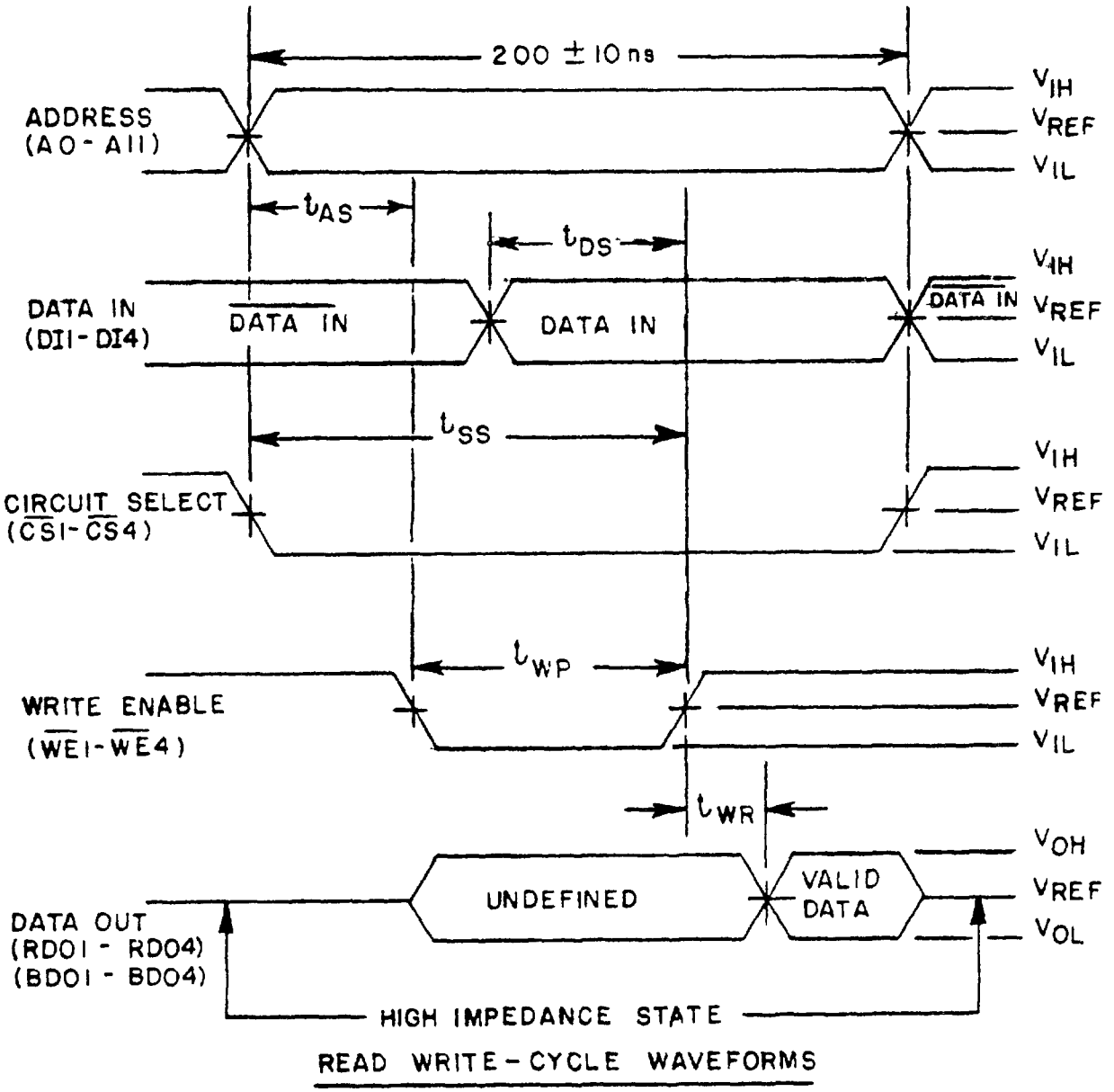
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:33Z
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CIRCUIT SELECT/DESELECT PATTERN PROGRAM FLOW CHART

FIGURE 25. Marching write recovery-one, marching write recovery-zero, and circuit select/deselect and chip independence pattern program flow charts and waveforms for table XVI - Continued.

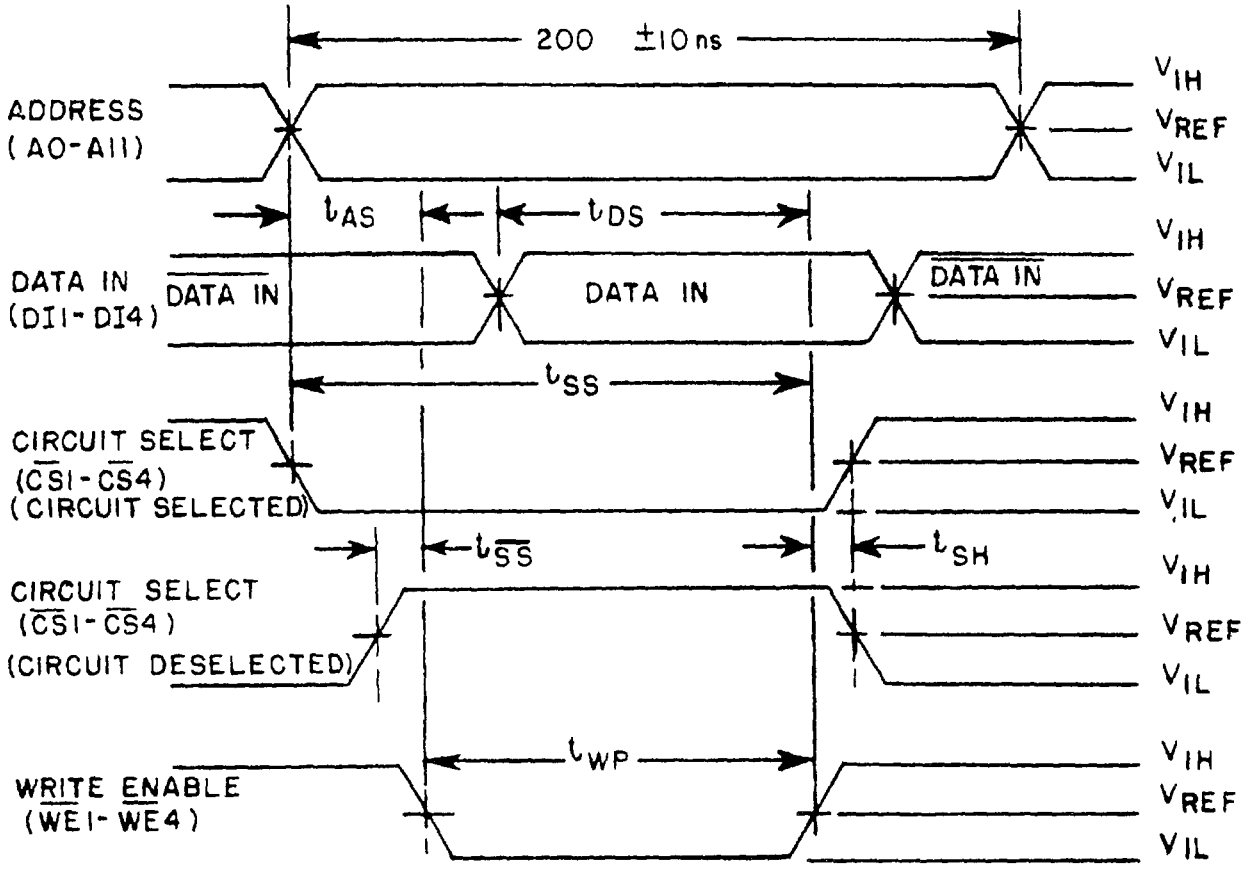
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:33Z
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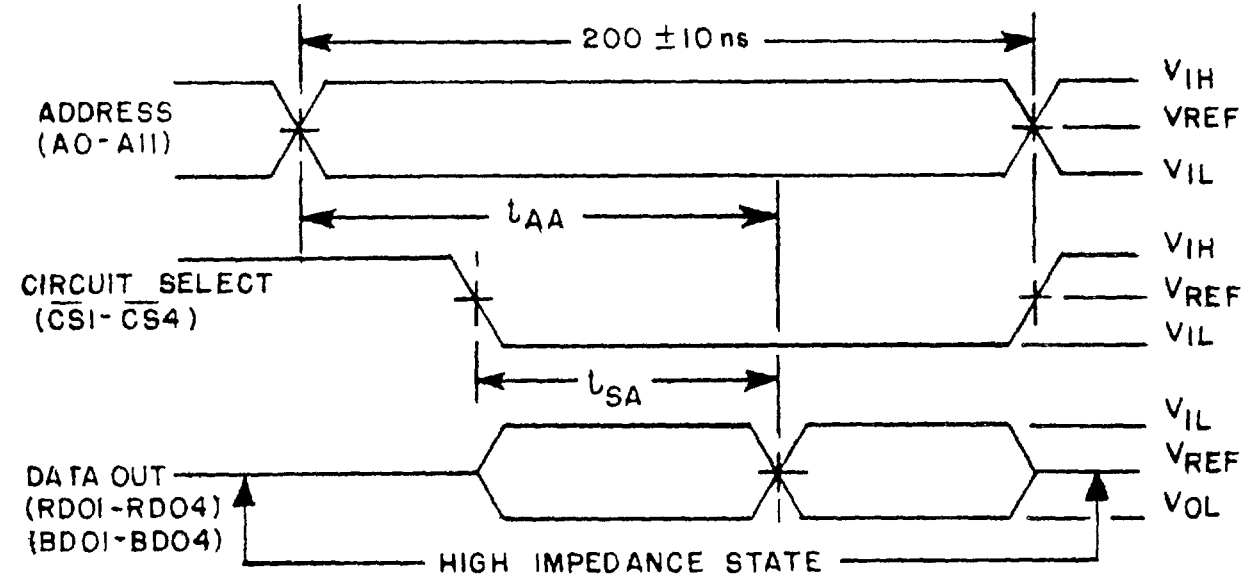
READ WRITE-CYCLE WAVEFORMS

FIGURE 25. Marching write recovery-one, marching write recovery-zero, and circuit select/deselect and chip independence pattern program flow charts and waveforms for table XVI - Continued.

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WRITE CYCLE WAVEFORMS



READ CYCLE WAVEFORMS

FIGURE 25. Marching write recovery-one, marching write recovery-zero, and circuit select/deselect and chip independence pattern program flow charts and waveforms for table XVI - Continued.

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NOTES:

1. The objective of the marching write recovery-one, marching write recovery-zero test pattern is to check write recovery time of the module at the same address. The objective of the circuit select deselect test pattern is to check for proper functioning of the address and circuit select circuitry in a dynamic environment. The patterns also verify address setup time (t_{AS}), data setup time (t_{DS}), circuit select setup time (t_{SS}), deselect setup time ($t_{\overline{SS}}$), circuit select hold time (t_{SH}), write recovery time (t_{WR}), write pulse width (t_{WP}), address access time (t_{AA}), and circuit setup access time (t_{SA}).
2. The chip independence test is a data pattern that is run during the marching write recovery test and the circuit select/deselect test. This pattern checks that writing to one data bit does not affect other data bits. To run this pattern, the entire marching write recovery-one, marching write recovery-zero and circuit select deselect patterns are run on each of the following data patterns and their complements.

	DI4	DI3	DI2	DI1
a.	0	0	0	1
b.	0	0	1	0
c.	0	1	0	0
d.	1	0	0	0

3. A program that fulfills the marching write recovery-one marching write recovery-zero pattern requirements would·

   a. Write and read pattern A at all memory address locations
   b. Read pattern A at address location zero.
   c. Write, then read pattern $\overline{A}$ at address location zero.
   d. Repeat (b) and (c) for address locations 1 through 4095, in turn.
   e. Set memory address location to zero.
   f. Read pattern $\overline{A}$ at location zero.
   g. Write, then read pattern A at address location zero.
   h. Repeat (f) and (g) for address locations 1 through 4095, in turn.
   i. Repeat steps (a) through (h) with patterns B, C, and D.
4. A program that fulfills the circuit select/deselect test pattern requirements would

   a. Write and read pattern A at all memory address locations.
   b Set memory address to location zero.
   c Decrement memory address by one, then read pattern A at maximum address.
   d. Increment memory address location by one
   e. Decrement memory address location by one, then, with select deselected, attempt to write pattern $\overline{A}$ at the memory address location
   f. Read pattern A at memory address location
   g. Decrement address location by one and repeat (d) through (f) for the resultant address location.
   h. Repeat (g) until reading address location zero.
   i. Repeat steps (a) through (h) pattern A and pattern $\overline{A}$ interchanged.
   j. Repeat steps (a) through (i) with patterns B, C, and D.

FIGURE 25. Marching write recovery-one, marching write recovery-zero, and circuit select/deselect and chip independence pattern program flow charts and waveforms for table XVI.

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Check the source to verify that this is the current version before use.

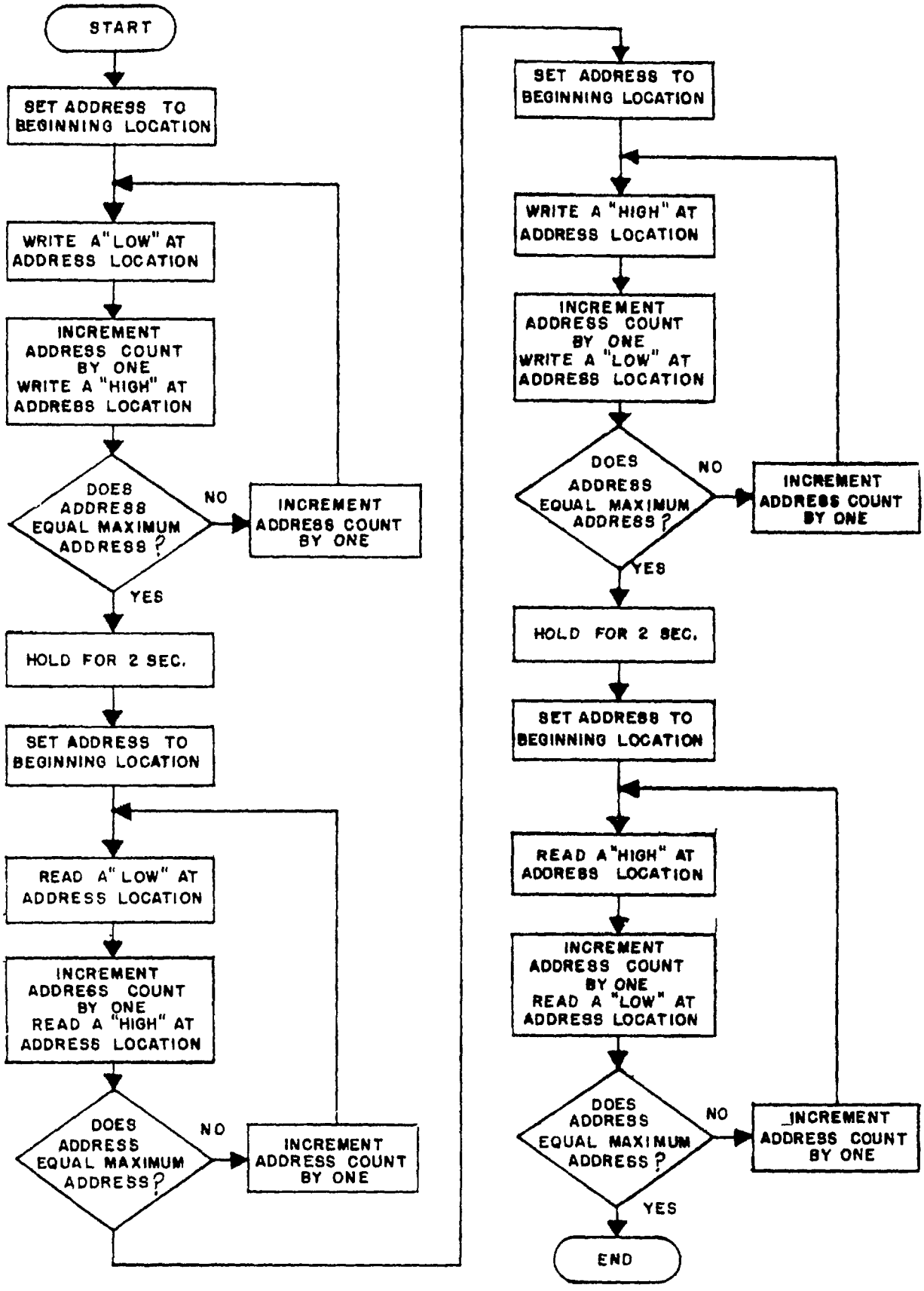
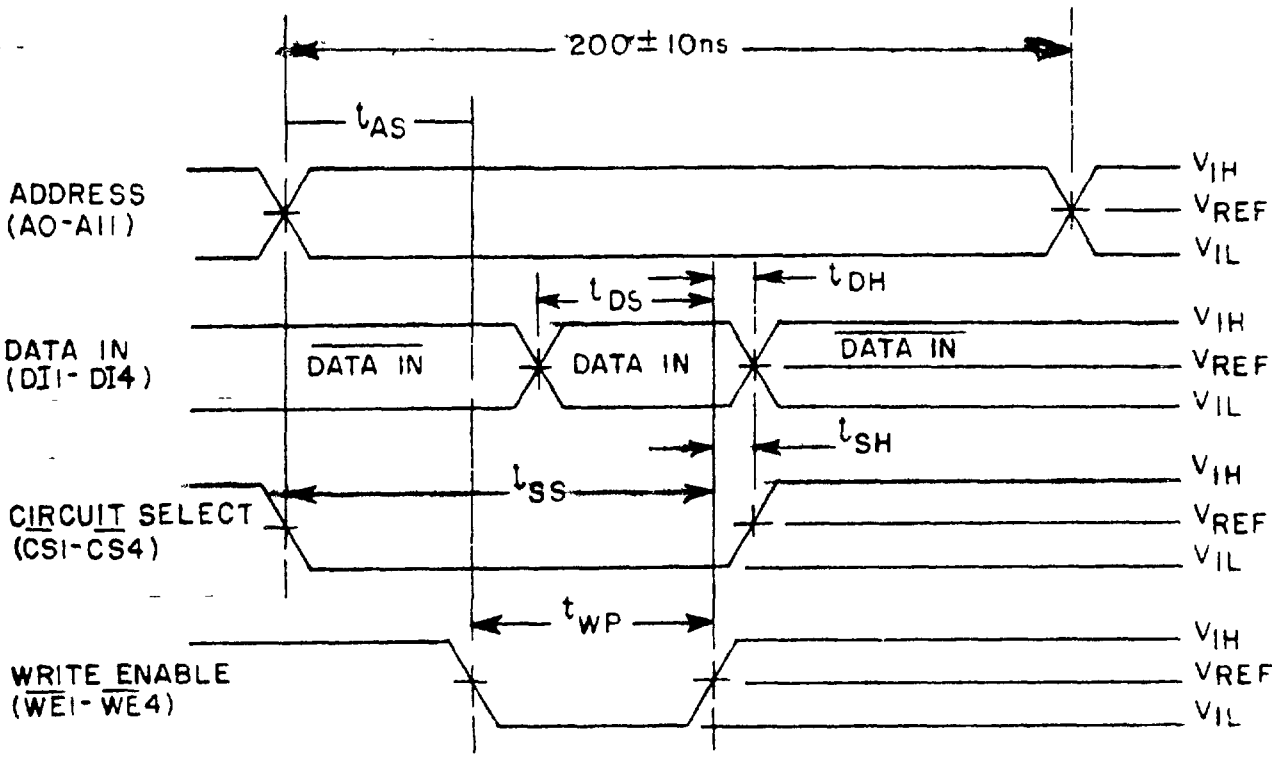
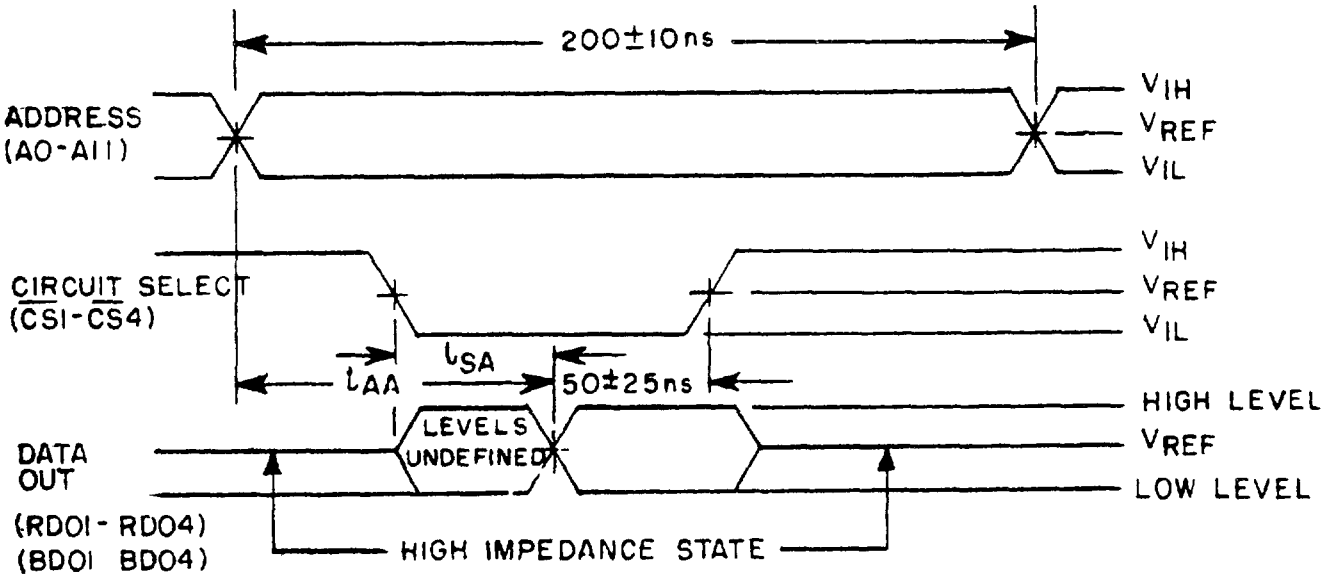


FIGURE 26. Static hold pattern program flow chart and waveforms for table XVI.

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LEAD CYCLE INPUT AND OUTPUT WAVEFORMS

FIGURE 26. Static hold pattern program flow chart and waveforms for table XVI - Continued.

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NOTES

1. The objective of the static hold test pattern is to check data retention after a static hold period of 2 seconds The pattern also verifies address setup time (t_{AS}), data setup time (t_{DS}), circuit select setup time (t_{SS}), data hold time (t_{DH}), circuit select hold time (t_{SH}), write pulse width (t_{WP}), address access time (t_{AA}), and circuit select access time (t_{SA}).
2. A pattern that fulfills the static hold test pattern requirements would:

   a Write a "low" at address location zero.
   b Write a "high" at address location one.
   c Repeat (a) and (b) for address locations 2 through 4095.
   d Hold the memory (no transitions) for 2 seconds.
   e. Read a "low" at address location zero.
   f. Read a "high" at address location one.
   g Repeat (e) and (f) for address locations 2 through 4095
   h Write a "high" at address location zero
   i Write a "low" at address location one.
   j. Repeat (h) and (i) for address locations 2 through 4095.
   k. Hold the memory (no transitions) for 2 seconds.
   l. Read a "high" at address location zero
   m Read a "low" at address location one.
   n. Repeat (l) and (m) for address locations 2 through 4095.
3. All address lines need not switch simultaneously each cycle

FIGURE 26. Static hold pattern program flow chart and waveforms for table XVI - Continued.

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Check the source to verify that this is the current version before use.

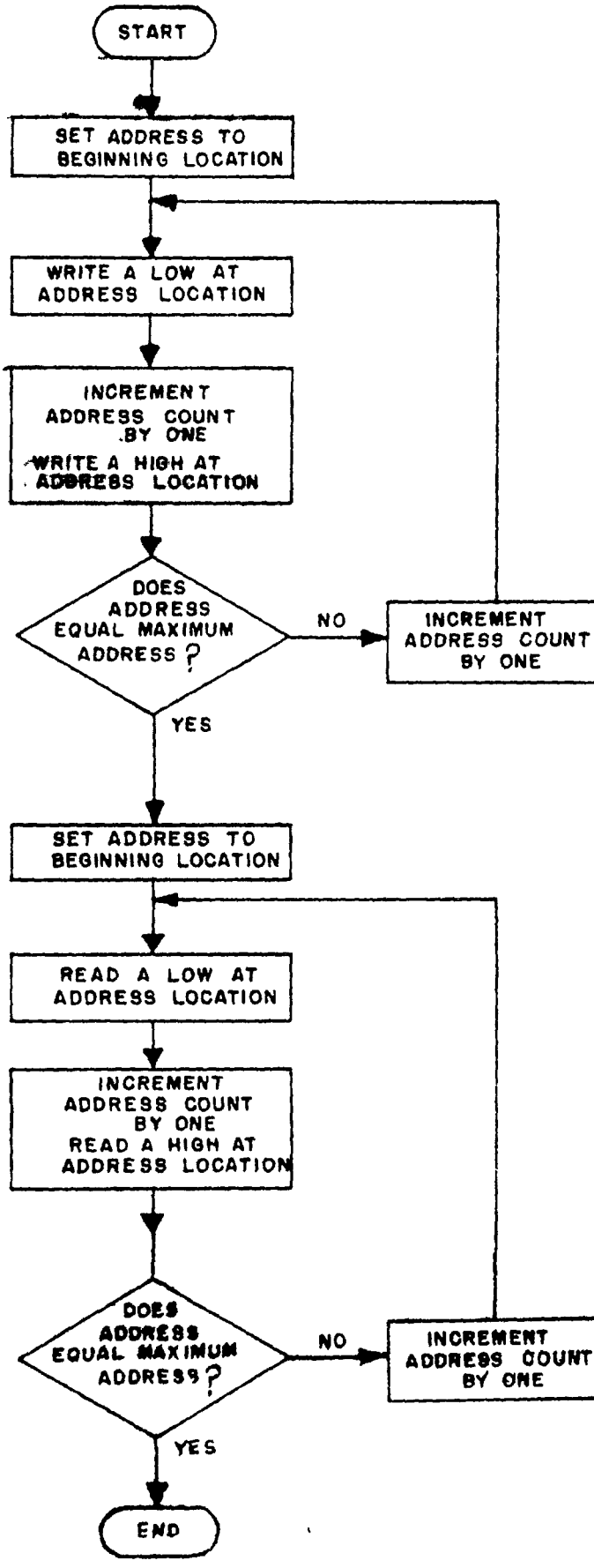
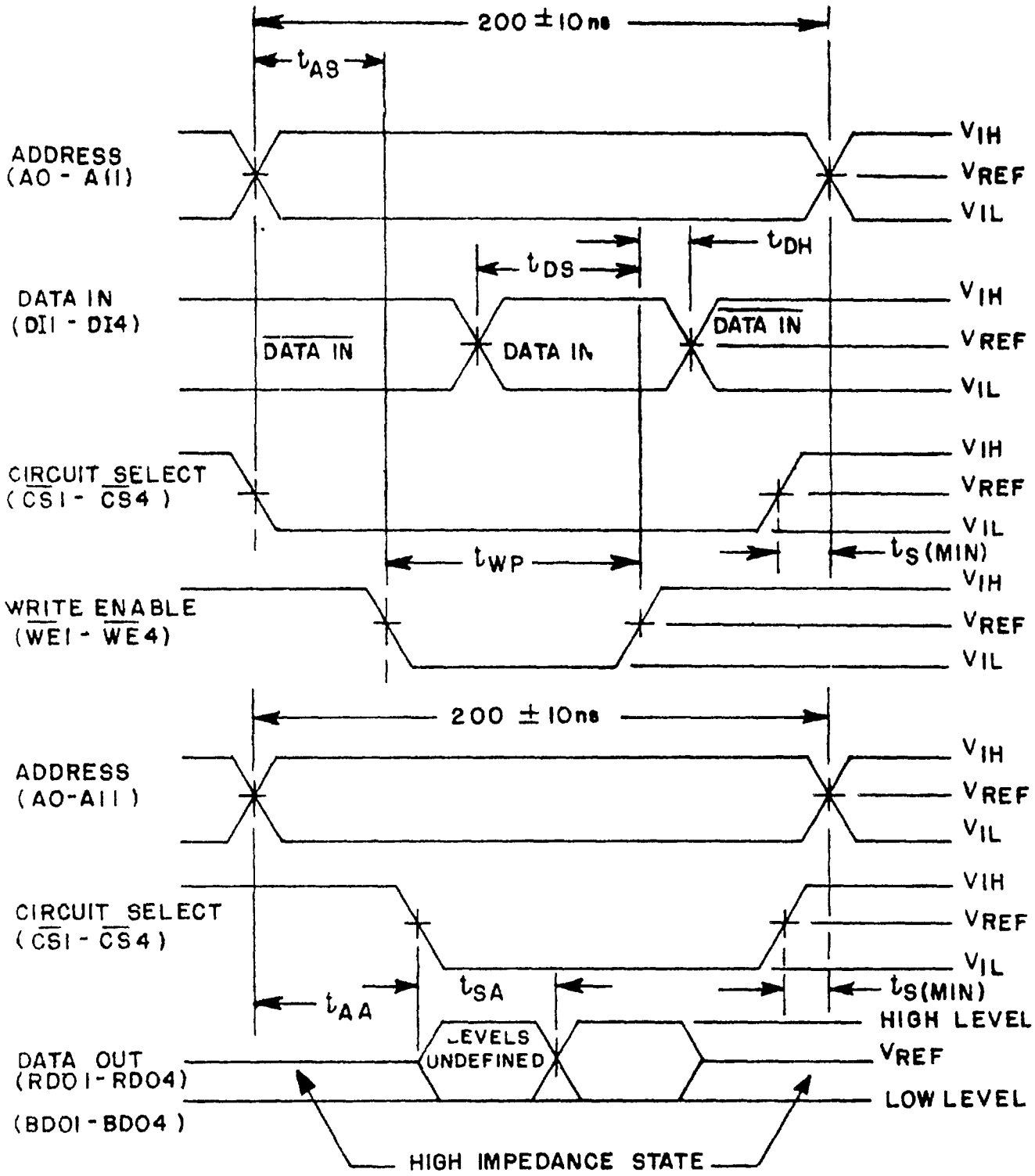


FIGURE 27. Minimum deselect time pattern program flow chart and waveforms for table XVI.

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LEAD CYCLE INPUT AND OUTPUT WAVEFORMS

FIGURE 27. Minimum deselect time pattern program flow chart and waveforms for table XVI - Continued.

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NOTES:
1. The objective of the minimum deselect time test pattern is to check circuit access time (t_{SA}) with a minimum deselect time. The pattern also checks address setup time (t_{AS}), data setup time (t_{DS}), circuit select setup time (t_{SS}), minimum write pulse width (t_{WP}), and data hold time (t_{DH}).
2. A pattern that fulfills the minimum deselect time test pattern requirements would:

   a. Write a "low" at address location zero.
   b. Write a "high" at address location one.
   c. Repeat a and b for address locations 2 through 4095.
   d. Read a "low" at address location zero.
   e. Read a "high" at address location one.
   f. Repeat d and e for address locations 2 through 4095.

3. All address lines need not switch simultaneously each cycle.

FIGURE 27. Minimum deselect time pattern program flow chart and waveforms for table XVI - Continued.

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TABLE XVII. Circuit select recovery time (t_{SRL} and t_{SRH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	$\overline{CS1}$ Circuit select recovery time (t_{SRL})	6,8,17, 18,19, 25,27, 29,30, 31,33	4	21	24	5	40	ns	28,29
S2	$\overline{CS2}$	4,6,8, 17,18, 19,27, 29,30, 31,33	25	28					
S3	$\overline{CS3}$	4,6,8, 17,18, 19,25, 27,29, 30,31	33	9					
S4	$\overline{CS4}$	4,6,8, 17,19, 25,27, 29,30, 31,33	18	11					
S5	$\overline{OE1}$	6,8,17, 18,19, 25,29, 30,31, 33	4, 27	22					
S6	$\overline{OE2}$	4,6,17, 18,19, 27,29, 30,31, 33	25, 8	26					
S7	$\overline{OE3}$	4,6,8, 17,18, 19,25, 27,30, 31	33, 29	13					

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TABLE XVII. Circuit select recovery time (t_{SRL} and t_{SRH}) test - Continued.

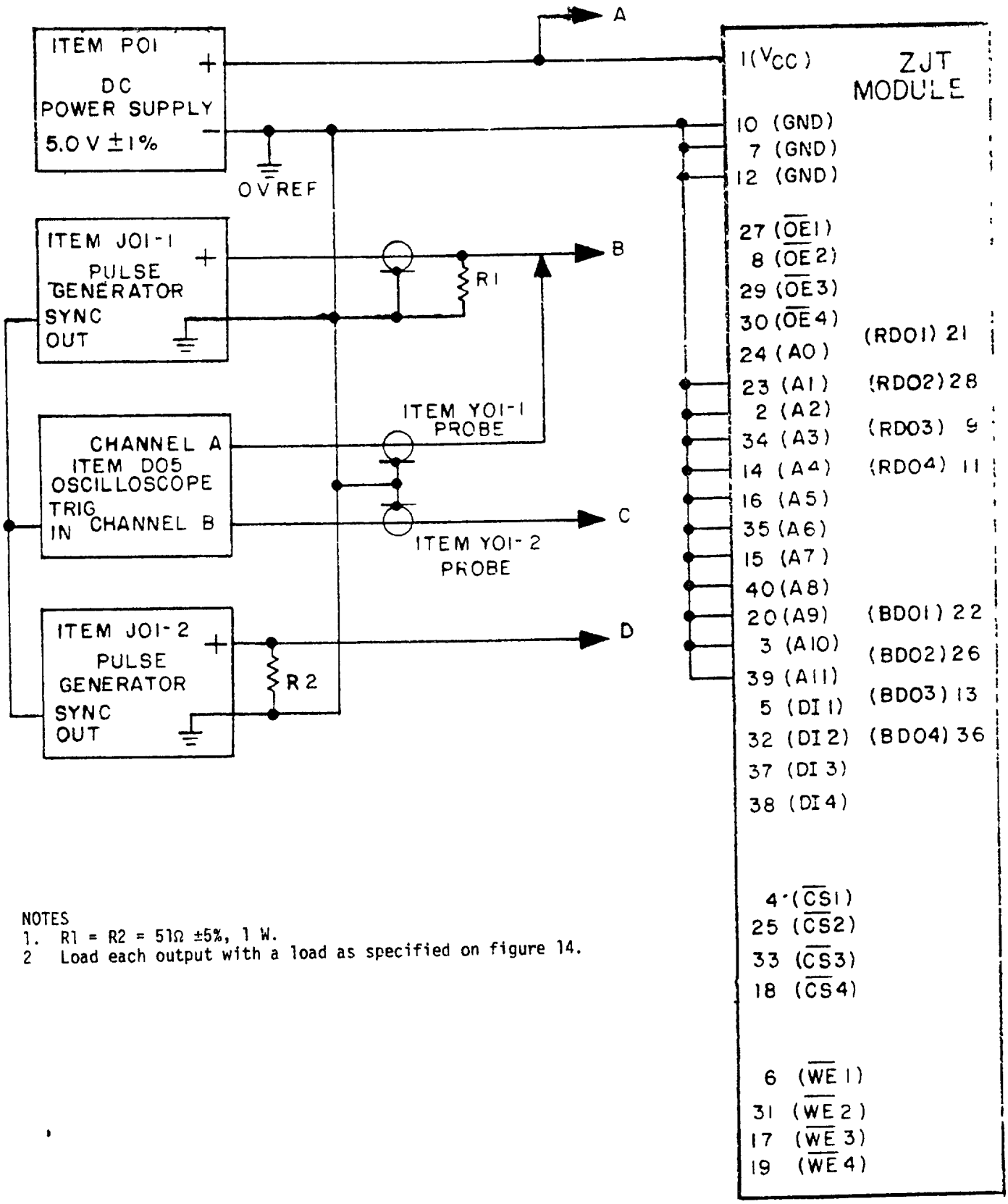
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S8	$\overline{OE}4$ Circuit select recovery time (t_{SRL})	4,6,8, 17,19, 25,27, 29,31, 33	18,30	36	24	5	40	ns	28,29
S9	$\overline{CS}1$ Circuit select recovery time (t_{SRH})	6,8,17, 18,19, 25,27, 29,30, 31,33	4	21					
S10	$\overline{CS}2$	4,6,8, 17,18, 19,27, 29,30, 31,33	25	28					
S11	$\overline{CS}3$	4,6,8, 17,18, 19,25, 27,29, 30,31	33	9					
S12	$\overline{CS}4$	4,6,8, 17,19, 25,27, 29,30, 31,33	18	11					
S13	$\overline{OE}1$	6,8,17, 18,19, 25,29, 30,31, 33	4,27	22					

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TABLE XVII. Circuit select recovery time (t_{SRL} and t_{SRH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S14	$\overline{OE}2$ Circuit select recovery time (t_{SRH})	4,6,17, 18,19, 27,29, 30,31, 33	25,8	26	24	5	40	ns	28,29
S15	$\overline{OE}3$ Circuit select recovery time (t_{SRH})	4,6,8, 17,18, 19,25, 27,30, 31	33,29	13	24	5	40	ns	28,29
S16	$\overline{OE}4$ Circuit select recovery time (t_{SRH})	4,6,8, 17,19, 25,27, 29,31, 33	18,30	36	24	5	40	ns	28,29

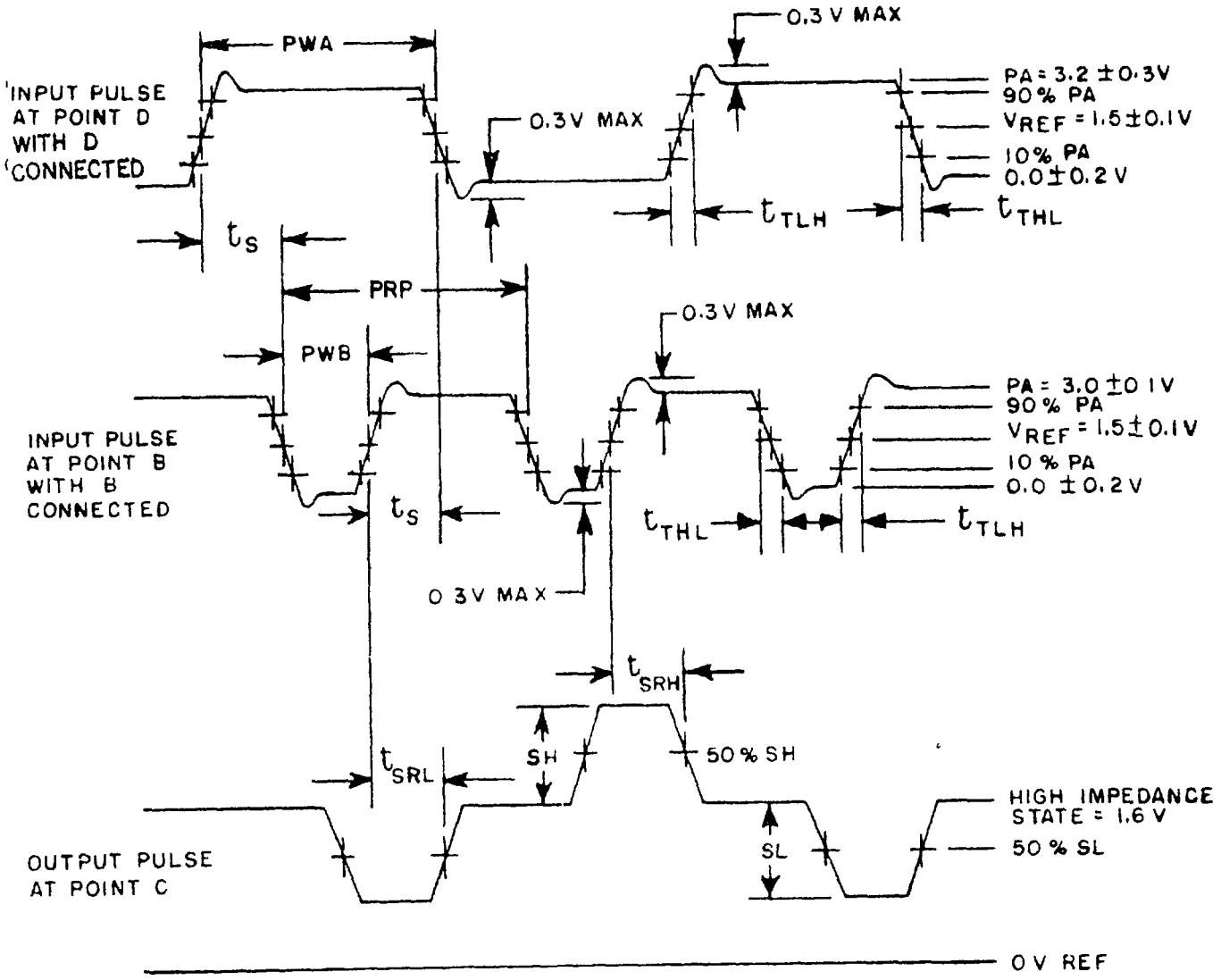
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:33Z
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NOTES
1. R1 = R2 = 51Ω ±5%, 1 W.
2 Load each output with a load as specified on figure 14.

FIGURE 28. Circuit select recovery time (t_{SRL} and t_{SRH}) test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:33Z
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NOTES:
1. Input pulse parameters:
   PWA = 2000 ±100 ns.
   PWB = 600 ±50 ns.
   PRP = 2000 ±100 ns.
   t_S = 700 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Initial setup procedure
   a. Store a "high" level in address location zero by connecting pins 24, 4, 18, 25 and 33 to pin 10 and connecting pins 5, 32, 37, and 38 to pin 1. Connect pins 6, 17, 19 and 31 to pin 10 for 400 ns minimum, then connect to point A.
   b. Store a "low" level in address location one by connecting pin 24 to pin 1 and connecting pins 4, 5, 18, 25, 32, 33, 37 and 38 to pin 10. Connect pins 6, 17, 19, and 31 to pin 10 for 400 ns minimum, then connect to point A.
   c. Verify that address location zero has a "high" stored and location one has a "low" stored.

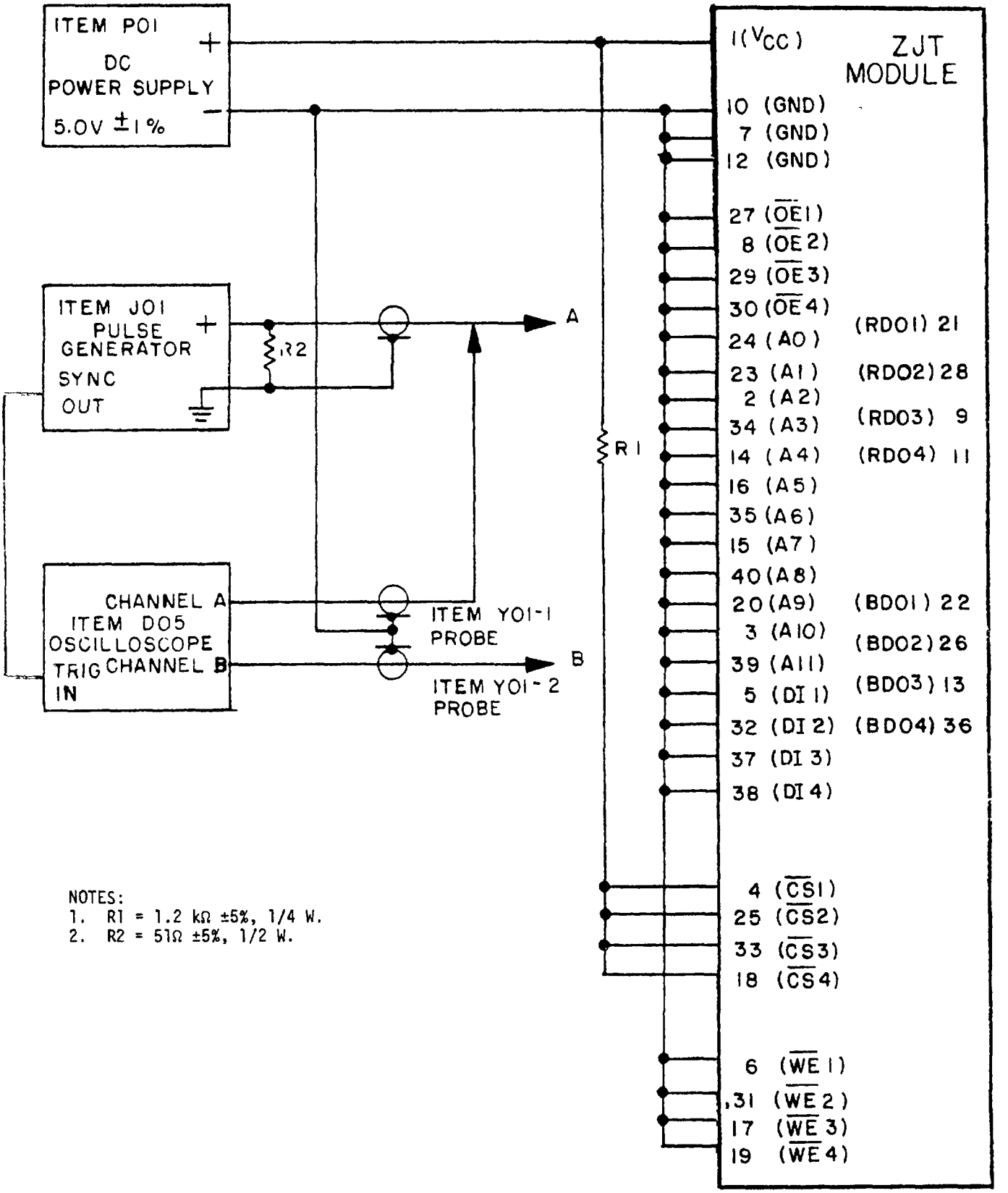
FIGURE 29. Circuit select recovery time (t_{SRL} and t_{SRH}) test circuit waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:33Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
T1	BD01	Propagation delay time (t_{PLH}), C_L = 50 pF	21	22	5	18	ns	30,31
T2	BD02		28	26				
T3	BD03		9	13				
T4	BD04		11	36				
T5	BD01	Propagation delay time (t_{PLH}), C_L = 150 pF	21	22		28		
T6	BD02		28	26				
T7	BD03		9	13				
T8	BD04		11	36				
T9	BD01	Propagation delay time (t_{PHL}), C_L = 50 pF	21	22		18		
T10	BD02		28	26				
T11	BD03		9	13				
T12	BD04		11	36				
T13	BD01	Propagation delay time (t_{PHL}), C_L = 150 pF	21	22		28		
T14	BD02		28	26				
T15	BD03		9	13				
T16	BD04		11	36				

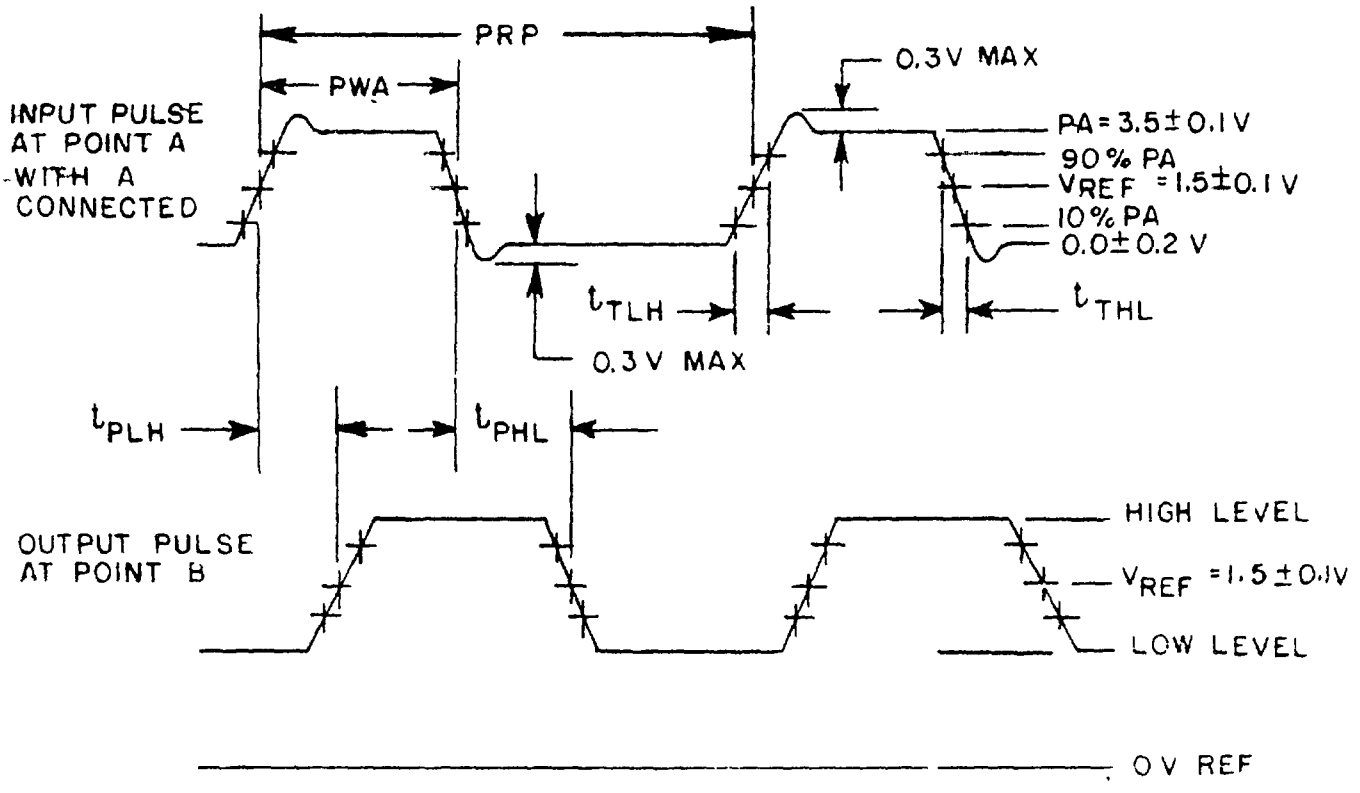
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:33Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1.2 kΩ ±5%, 1/4 W.
2. R2 = 51Ω ±5%, 1/2 W.

FIGURE 30. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVIII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 1000 ±100 ns.
   PRP = 2000 ±200 ns.
2. Before making the measurement verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V.
3. All input voltages are referenced to ground.

FIGURE 31. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVIII.

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TABLE XIX. Input crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	$\overline{CS1}$ Input crosstalk, low level	6,17,18,19,25, 31,33	4	0	500	mV	32,33
U2	$\overline{CS2}$	4,6,17,18,19, 31,33	25				
U3	$\overline{CS3}$	4,6,17,18,19, 25,31	33				
U4	$\overline{CS4}$	4,6,17,19,25, 31,33	18				
U5	$\overline{WE1}$	4,17,18,19,25, 31,33	6				
U6	$\overline{WE2}$	4,6,17,18,19, 25,33	31				
U7	$\overline{WE3}$	4,6,18,19,25, 31,33	17				
U8	$\overline{WE4}$	4,6,17,18,25, 31,33	19				
U9	$\overline{CS1}$ Input crosstalk, high level	6,17,18,19,25, 31,33	4				
U10	$\overline{CS2}$	4,6,17,18,19, 31,33	25				
U11	$\overline{CS3}$	4,6,17,18,19, 25,31	33				
U12	$\overline{CS4}$	4,6,17,19,25, 31,33	18				

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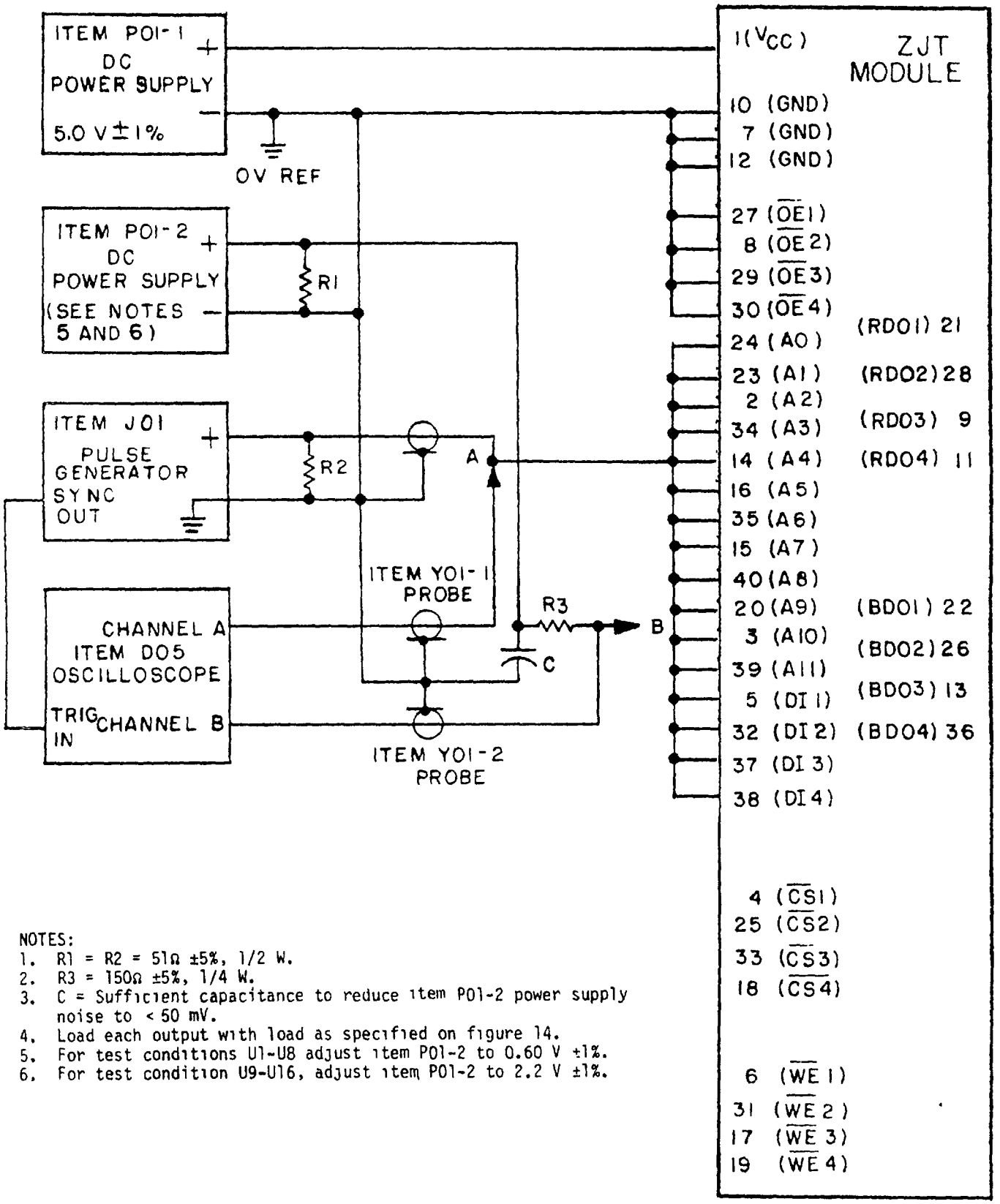
TABLE XIX. Input crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U13	$\overline{WE1}$ Input crosstalk, high level	4,17,18,19,25, 31,33	6	0	500	mV	32,33
U14	$\overline{WE2}$	4,6,17,18,19, 25,33	31				
U15	$\overline{WE3}$	4,6,18,19,25, 31,33	17				
U16	$\overline{WE4}$	4,6,17,18,25, 31,33	19				

TABLE XX. Power supply transient amplitude and ripple amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
V1	Power supply transient amplitude	0	600	mV	34,35
V2	Power supply ripple amplitude	0	700	mV	34,35

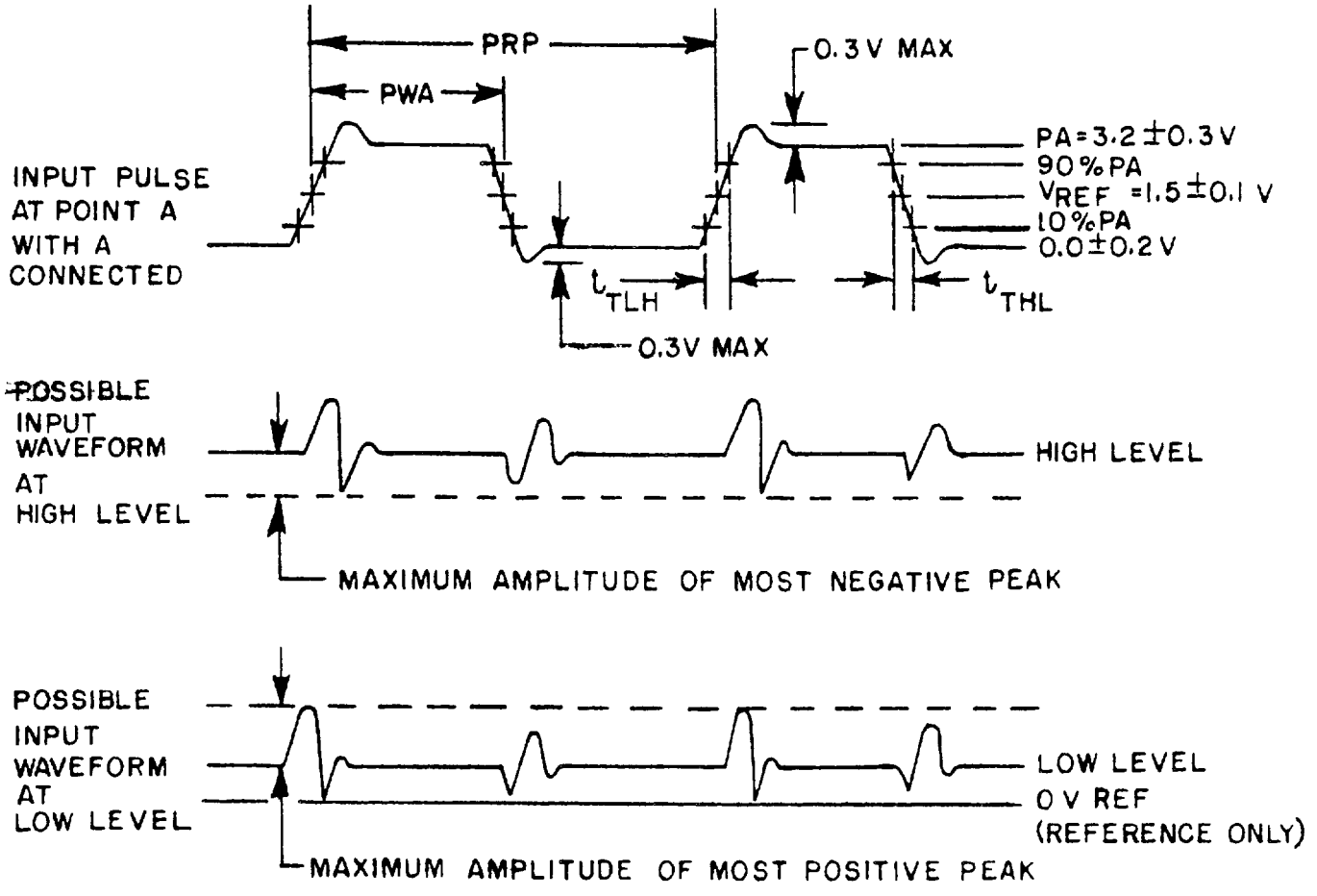
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 150Ω ±5%, 1/4 W.
3. C = Sufficient capacitance to reduce item P01-2 power supply noise to < 50 mV.
4. Load each output with load as specified on figure 14.
5. For test conditions U1-U8 adjust item P01-2 to 0.60 V ±1%.
6. For test condition U9-U16, adjust item P01-2 to 2.2 V ±1%.

FIGURE 32. Input crosstalk (low-level and high-level) test circuit for table XIX.

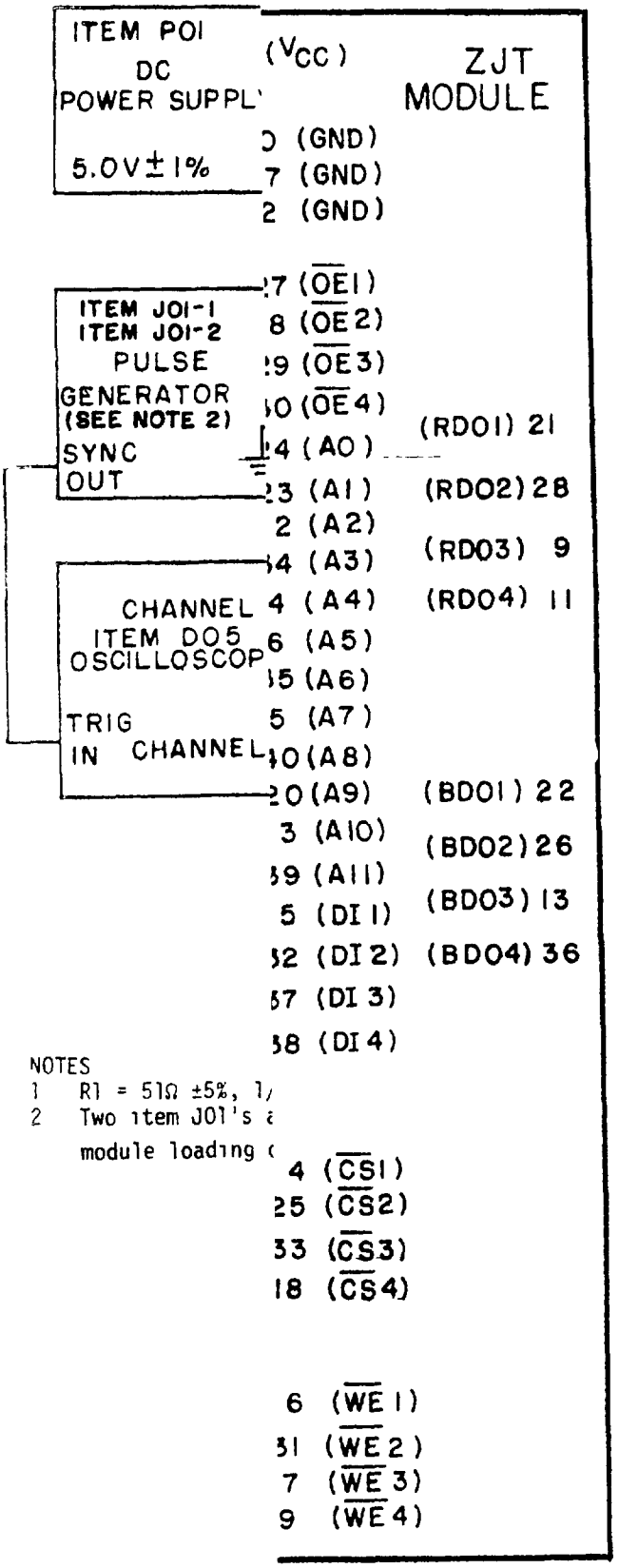
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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 20 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
2. The lead length of electrical conductors between terminator resistor R2 and input pins, probes Y01-1, Y01-2 to input pins, load circuits to output pins, and ground leads of probes Y01-1, Y01-2 to test fixture and module ground pins shall be less than two inches (50.8 mm).

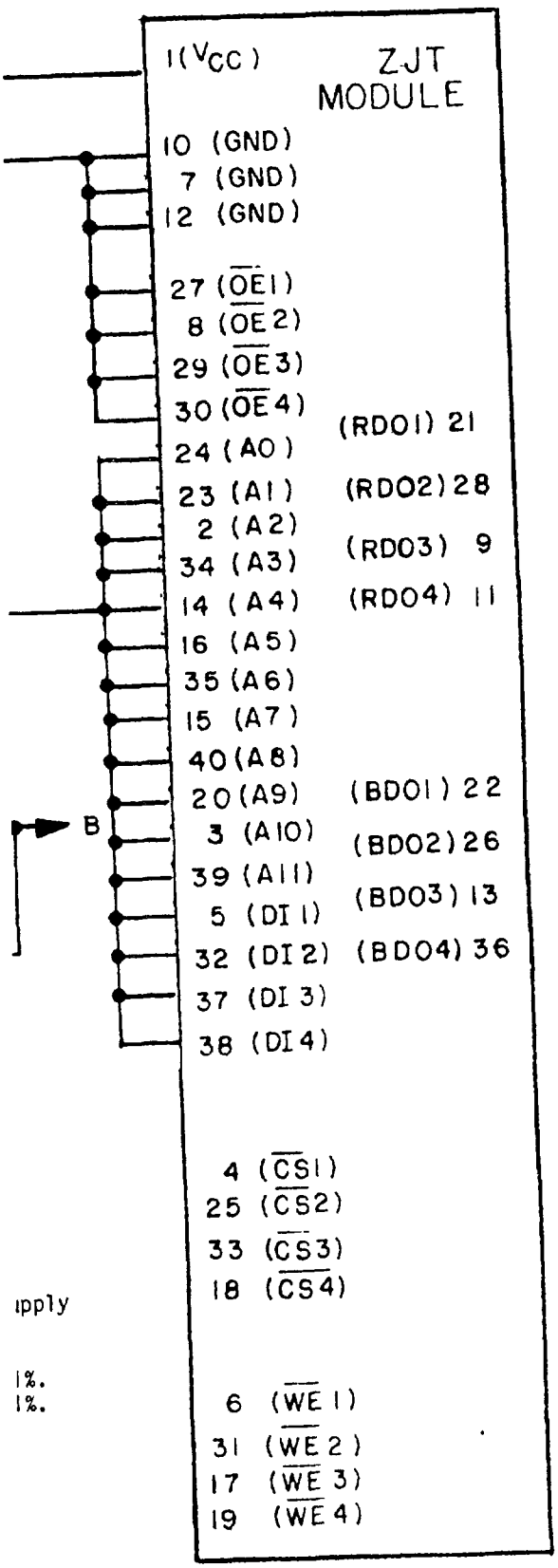
FIGURE 33. Input crosstalk (low-level and high-level) test circuit waveforms for table XIX.

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NOTES
1 R1 = 51Ω ±5%, 1/
2 Two item JO1's a
module loading (

FIGURE 34. Pow ircuit for table XX.

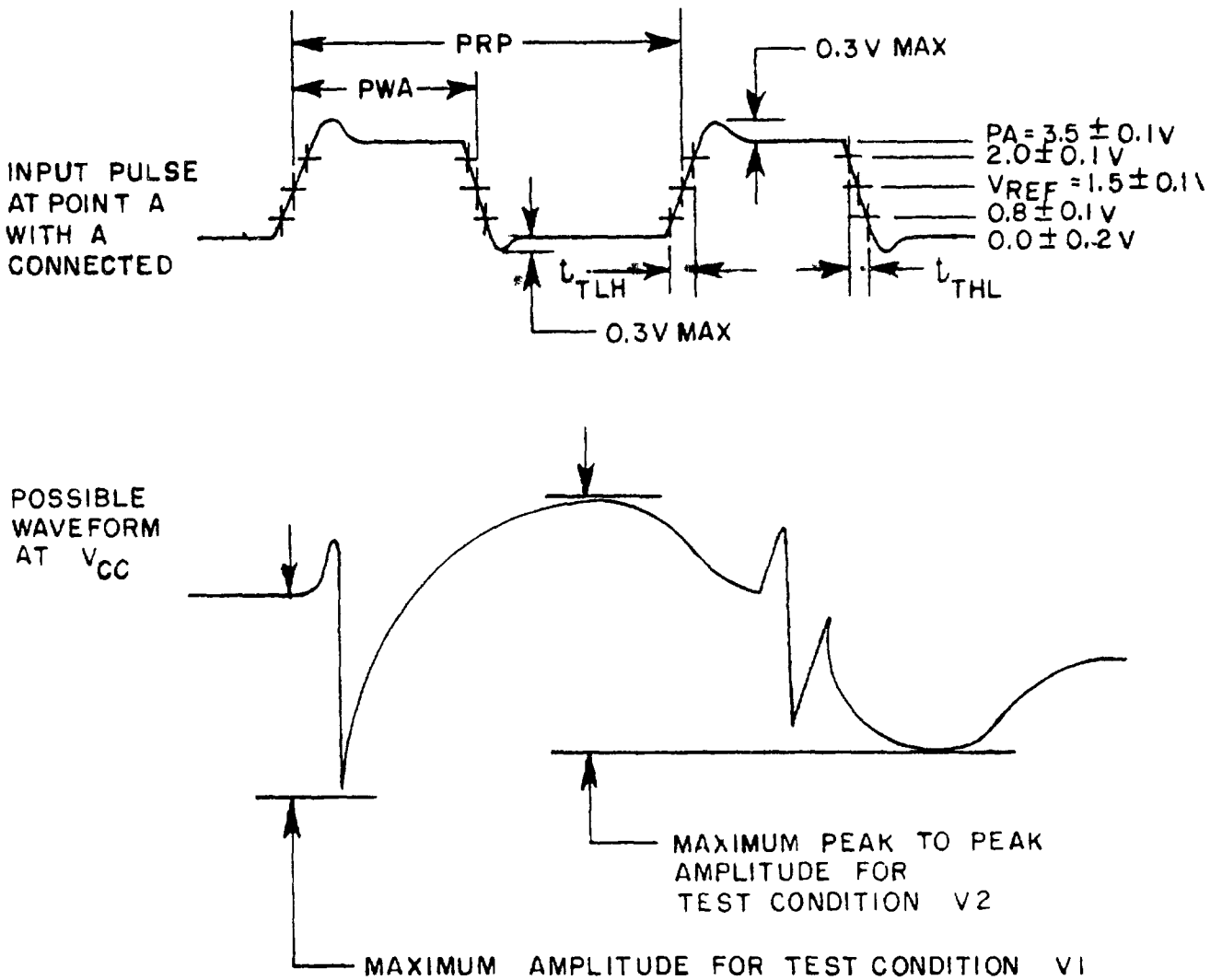


ıpply

1%.
1%.

1) test circuit for table XIX

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NOTES:
1. Input pulse parameters for items J01-1 and J01-2.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 3000 ±300 ns.
2. All voltages are referenced to ground.

FIGURE 35. Power supply transient amplitude and ripple amplitude test circuit waveforms for table XX.

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TABLE XXI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W	Power supply filter capacitance	Open	1	6	10	µF	36

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TABLE XXII. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	$\overline{CS1}$ Input capacitance	4	Open	2	15	pF	36
X2	$\overline{CS2}$	25					
X3	$\overline{CS3}$	33					
X4	$\overline{CS4}$	18					
X5	A0	24			45		
X6	A1	23					
X7	A2	2					
X8	A3	34					
X9	A4	14					
X10	A5	16					
X11	A6	35					
X12	A7	15					
X13	A8	40					
X14	A9	20					
X15	A10	3					
X16	A11	39			15		
X17	$\overline{WE1}$	6			15		
X18	$\overline{WE2}$	31			15		

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TABLE XXII. Input capacitance test - Continued.

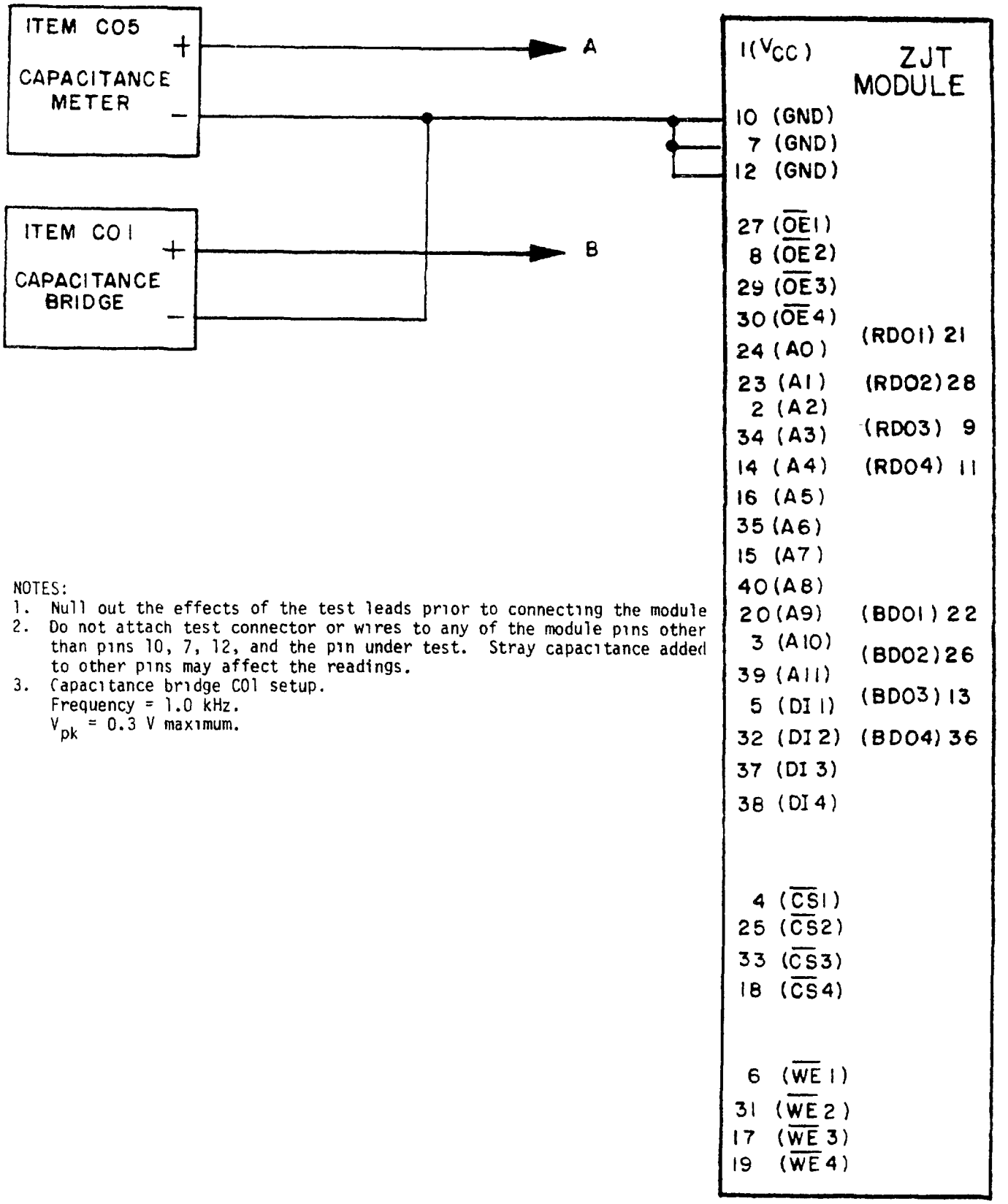
Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
X19	$\overline{WE3}$	Input capacitance	17	Open	2	15	pF	36
X20	$\overline{WE4}$		19					
X21	DI1		5					
X22	DI2		32					
X23	DI3		37					
X24	DI4		38					
X25	$\overline{OE1}$		27					
X26	$\overline{OE2}$		8					
X27	$\overline{OE3}$		29					
X28	$\overline{OE4}$		30					

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TABLE XXIII. Output capacitance test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Y1	BD01	Output capacitance	22	Open	2	25	pF	36
Y2	BD02		26					
Y3	BD03		13					
Y4	BD04		36					
Y5	RD01		21					
Y6	RD02		28					
Y7	RD03		9					
Y8	RD04		11					

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NOTES:
1. Null out the effects of the test leads prior to connecting the module
2. Do not attach test connector or wires to any of the module pins other than pins 10, 7, 12, and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge CO1 setup.
   Frequency = 1.0 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 36. Power supply filter, input, and output capacitance test circuit for tables XXI, XXII, and XXIII.

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